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Vehicle imaging and inspection system

Abstract

A front-end image acquisition component acquires photographs and/or videos of the exterior of a vehicle traveling on a path between a first predefined location and a second predefined location. In one aspect, an image acquisition system comprises a first camera support structure, a second camera support structure, an image processor, and a memory. Cameras are affixed to each of the camera support structures for acquiring image data of a subject vehicle. Image data is controlled by the image processor and the memory. In another aspect, an image acquisition system comprises a first camera tower, a second camera tower, a camera boom. Cameras are affixed to each of the camera towers.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/407,898 filed Jan. 9, 2024, which is a continuation of U.S. patent application Ser. No. 18/533,995 filed Dec. 8, 2023, which is continuation of U.S. patent application Ser. No. 18/341,504 filed Jun. 26, 2023, now allowed, which is a continuation of U.S. patent application Ser. No. 17/574,025 filed Jan. 12, 2022, now U.S. patent Ser. No. 11,756,110 issued Sep. 12, 2023, which is a continuation-in-part of U.S. patent application Ser. No. 16/385,329 filed Apr. 16, 2019, abandoned, which is a continuation-in-part of U.S. patent application Ser. No. 15/012,033 filed Feb. 1, 2016, now U.S. Pat. No. 10,681,261 issued Jun. 9, 2020, which is a continuation-in-part of U.S. patent application Ser. No. 14/574,638 filed Dec. 18, 2014, abandoned, which is a continuation-in-part of U.S. patent application Ser. No. 14/088,939 filed Nov. 25, 2013, abandoned, which claims the benefit of U.S. Provisional Patent Application No. 61/792,258 filed Mar. 15, 2013 and U.S. Provisional Patent Application No. 61/732,078 filed Nov. 30, 2012. U.S. patent application Ser. No. 16/385,329 is also a divisional of U.S. patent application Ser. No. 14/575,260 filed Dec. 18, 2014, now U.S. Pat. No. 10,269,059 issued Apr. 23, 2019, which is a continuation-in-part of U.S. patent application Ser. No. 14/574,638 filed Dec. 18, 2014, abandoned. U.S. patent application Ser. No. 14/575,260 also claims the benefit of U.S. Provisional Patent Application No. 61/920,576 filed Dec. 24, 2013 and U.S. Provisional Patent Application No. 61/918,492 filed Dec. 19, 2013. U.S. patent application Ser. No. 17/574,025 is also a continuation-in-part of U.S. patent application Ser. No. 15/945,828 filed Apr. 5, 2018, now U.S. Pat. No. 11,270,350 issued Mar. 8, 2022, which claims the benefit of U.S. Provisional Patent Application No. 62/484,197 filed Apr. 11, 2017. U.S. patent application Ser. No. 15/945,828 is also a continuation-in-part of U.S. patent application Ser. No. 14/575,260. U.S. patent application Ser. No. 15/945,828 is also a continuation-in-part of U.S. patent application Ser. No. 15/012,033. The

entire contents of the above applications and patents are incorporated herein by reference for all purposes.

BACKGROUND

(1) Consumers usually make purchases in one of two ways. The first, a traditional approach, involves visiting a dealer or merchant's place of business, listening to a salesperson's pitch, testing selected goods, and then haggling over price. Similarly, consumers purchase real estate by the traditional approach by visiting and touring the real estate and then haggling over price. This approach advantageously provides consumers an opportunity to view the particular product or real estate in-person and receive hands-on demonstrations of features and options or the ability to view it in its entirety. However, this approach is time-consuming and interacting with a salesperson can be intimidating for many consumers.

(2) The second approach involves a “virtual showroom” where consumers can research and price goods, real estate, and the like on the Internet. This approach advantageously allows consumers to search for goods or real estate in less time and with less salesperson interaction than the traditional approach. Unfortunately, when visiting a virtual showroom, consumers are often limited to stock photographs and generic information and must forego the more traditional “walk around” experience. Current systems and methods of acquiring photographs and/or videos of goods or real estate and making them available on the Internet are labor-intensive and require skillful use of technologically advanced devices and software. Consequently, even if actual photographs and/or videos of a particular product, real estate, or the like are available, they are often limited in scope and cannot adequately convey how a product or real estate appears in-person and its actual condition. In addition, the information available is often non-specific and relates generally to a make and model rather than specifically to the particular item. In other words, consumers sacrifice advantageous aspects of the traditional goods or real estate purchasing approach in order to enjoy the convenience and other advantages of the virtual showroom approach.

(3) Moreover, many virtual sellers often employ a “no haggle” pricing model. As a result, it becomes important for virtual sellers to make a high volume of sales. A virtual seller attempting to make a high volume of sales must quickly acquire information about goods or real estate in its inventory, including photographs and/or videos of the goods or real estate, and make that information available to consumers on the Internet. Conventional systems and methods are unable to acquire photographs and/or videos of goods or real estate and post them on the Internet with the necessary throughput to keep up with the quick turnaround of high-volume sales.

SUMMARY

(4) In accordance with an embodiment of the invention, the front-end of an automated studio collects images and/or videos of items under inspection, such as vehicles, goods, real estate, or the like and makes them available on the Internet. An aspect of this end-to-end solution captures images and other information of an item under inspection, processes and stores the images and information, and adds the item to a virtual showroom or other virtual inspection site that presents an inventory of such vehicles, goods, real estate, or the like. This aspect of the solution provides users with ample information about the particular item under inspection. For example, users, such as consumers, can inspect vehicles available in the inventory virtually along with the traditional benefits of a showroom. Further, this aspect of the solution provides sellers with a user-friendly and quick way of acquiring images and information about the items and creating a virtual showroom to display them (e.g., 7 minutes or less per item under inspection). This aspect makes high-volume “no-haggle” sales tactics economical to employ. Also in accordance with aspects of the invention, image acquisition is automated in conjunction with the movement of an item under inspection from a first location to a second location. Additionally, in accordance with aspects of the invention, the processing of the imaged item can include the identification or detection of damage, defects, and/or other identifiable features with the execution of computer-executable functions.

- (5) In an aspect, a system for acquiring images of an item, such as a vehicle, includes a gantry structure that has a first leg, a second leg, a cross beam, a first plurality of side view cameras, a second plurality of side view cameras, and a processor. The first leg and the second leg are located between a first predefined location and a second predefined location on opposite sides of a travel path for the vehicle. The cross beam is located above the travel path and bridges the first and second legs. Each plurality of side view cameras is affixed to a corresponding leg (i.e., the first plurality of cameras is affixed to the first leg). The side view cameras are configured for acquiring image data of the exterior of the vehicle traveling on the path from the first predefined location to the second predefined location. The processor is configured for executing computer-executable functions for processing the images from one or more of the side view cameras.
- (6) In another aspect, a system for acquiring images of an item, such as a vehicle, includes a first camera tower, a second camera tower, a camera boom, a first plurality of side view cameras, and a second plurality of side view cameras. The first camera tower and the second camera tower are located between a first predefined location and a second predefined location on opposite sides of a travel path for the vehicle. The camera boom is affixed to the first camera tower and extends toward the second camera tower above the travel path. Each plurality of side view cameras is affixed to a corresponding camera tower (i.e., the first plurality of cameras is affixed to the first camera tower). The side view cameras are configured for acquiring image data of the exterior of the vehicle traveling on the path from the first predefined location to the second predefined location.
- (7) In yet another aspect, a method of identifying or detecting damage or defects in the exterior of an item, such as a vehicle, includes the steps of providing a front-end image acquisition component, a database, and a processor, acquiring a series of new images of the vehicle with the front-end image acquisition component, and using a processor to execute computer-executable functions. The front-end image acquisition component has a plurality of cameras configured for recording a series of new images of the vehicle. The database includes previously acquired images of the same or other vehicles and information associated with the previously acquired images. The processor is configured for executing computer-executable functions for processing the new images taken by one or more of the plurality of cameras. The acquisition of new images of the vehicle involves the activation of one or more of the plurality of cameras in the front-end image acquisition component as the vehicle moves on a path from a first predefined location to a second predefined location. The processor is configured to execute computer-executable functions that compare the new images taken by the front-end image acquisition component with the contents of the database to identify or detect damage or defects in the exterior of the vehicle.
- (8) Other objects and features will be in part apparent and in part pointed out hereinafter.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram of an automated studio comprised of various components in accordance with an embodiment of the invention.
- (2) FIG. 2A is a diagram of a mobile image acquisition apparatus comprised of various components for use in the automated studio of FIG. 1 in accordance with an embodiment of the invention.
- (3) FIG. 2B is a diagram of a tripod device for use with the mobile image acquisition apparatus of FIG. 2A.
- (4) FIGS. 3A-C illustrate a turntable and camera within an imaging space for use in the automated studio of FIG. 1 in accordance with an embodiment of the invention.
- (5) FIGS. 4A-B illustrate an imaging studio comprised of various components for use in the automated studio of FIG. 1 in accordance with an embodiment of the invention.

- (6) FIGS. 5A-C illustrate an imaging apparatus comprised of cameras, a support arm, and a linear motion member for use in the automated studio of FIG. 1 in accordance with an embodiment of the invention.
- (7) FIG. 5D illustrates an imaging apparatus comprised of various components for use in the automated studio of FIG. 1 in accordance with an embodiment of the invention.
- (8) FIGS. 6A-E illustrate an imaging studio comprised of cameras, camera towers, camera booms, and a vehicle transporter for use in the automated studio of FIG. 1 in accordance with an embodiment of the invention.
- (9) FIGS. 7A-7F illustrate an exemplary GUI for acquiring images and/or information and receiving commands from a user.
- (10) FIGS. 8A and 8B illustrate an imaging apparatus comprised of cameras, arms, and a hub for use in the automated studio of FIG. 1 in accordance with an embodiment of the invention.
- (11) FIGS. 9A-9D illustrate the imaging apparatus of FIGS. 8A and 8B with supports in accordance with an embodiment of the invention.
- (12) FIGS. 10A-10D and 11A-11D illustrate the imaging apparatus of FIGS. 8A and 8B with supports and a structure mount in accordance with an embodiment of the invention.
- (13) FIG. 12A is a block diagram illustrating a data structure for criterion identifiers and associated criterion values for an item of interest according to an embodiment of the invention.
- (14) FIG. 12B illustrates a component for determining criterion values associated with criterion identifiers for an item of interest according to an embodiment of the invention.
- (15) FIG. 12C is a flow chart illustrating a method of determining criterion values associated with criterion identifiers for an item of interest according to an embodiment of the invention.
- (16) FIG. 13A is a block diagram illustrating a data structure for criterion identifiers and associated criterion values for a desired item according to an embodiment of the invention.
- (17) FIG. 13B illustrates a component for determining when a criterion value for a desired item satisfies a criterion value of an item of interest according to an embodiment of the invention.
- (18) FIG. 13C is a flow chart illustrating a method of determining when a criterion value for a desired item satisfies a criterion value of an item of interest according to an embodiment of the invention.
- (19) FIG. 13D is a flow chart illustrating a method of determining when an amount of matched criterion values satisfies a threshold value according to an embodiment of the invention.
- (20) Corresponding reference characters indicate corresponding parts throughout the drawings.
- (21) Corresponding reference characters indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION OF THE INVENTION

- (22) FIG. 1 illustrates an automated studio, generally indicated at **100**, embodying aspects of the present invention. In accordance with aspects of the present invention, the automated studio **100** collects images and/or videos of objects or scenes and makes them available on the Internet.
- (23) In an embodiment, automated studio **100** includes a front-end image acquisition component **102** and a computer **103**. The studio **100** communicates via a server **104** to a database **105**, end-user console portals **106**, and administrator console portal **107**. The front-end image acquisition component **102** is communicatively connected to the computer **103**, which is communicatively connected to the server **104**. As shown in FIG. 1, server **104** is communicatively connected to the database **105**, which is communicatively connected to the consumer-side portal access **106** and the administrator console portal **107**.
- (24) The automated studio **100** provides an end-to-end solution in which images of a particular object are acquired, processed, stored, and used to create an entry for the object in an online inventory of objects, all with minimal human effort and in a short enough period of time to make a high volume of image acquisitions economically feasible. In an aspect, automated studio **100** provides simplicity and portability and employs software that can be utilized across multiple sites and platforms and is easy to use, install, and support. Contemplated objects include, by way of

example and not limitation, automobiles, boats, watercraft, recreational vehicles, motorcycles, all-terrain vehicles, trailers, aircraft, vessel interiors, real estate and associated structures and fixtures, and the like. Beneficially, automated studio **100** provides improvements in the fields of automated object imaging, online marketplaces, and transportation maintenance and safety by providing a means to acquire images of a particular object and upload them to an online inventory or database in a small amount of time and with little or no human involvement.

(25) According to aspects of the invention, front-end image acquisition component **102** is adapted for acquiring images of an object of interest. As used herein, images include still photographs (e.g., snap shots), moving photographs (e.g., videos or movies), panoramics, stereoscopic photographs, infrared images, or any combination thereof. In addition to acquiring images of an entire object, images of only a portion of an object may be acquired. For example, front-end image acquisition component **102** is configured for taking images of identifying information or any areas of the object that are particularly noteworthy, such as damaged areas, areas most likely to be damaged, identification numbers, and the like. It is also contemplated that a salesperson can provide a video demonstration of the object's features and options to be integrated into the final virtual showroom or inventory.

(26) Referring further to FIG. **1**, computer **103** executes computer-readable instructions embodied in software, namely, an automatic photography application that integrates with front-end image acquisition component **102**. In one embodiment, computer **103** displays, on a graphical user interface (GUI), a systematic process for acquiring images of an object and receives commands from a user to acquire the images. For example, computer **103** may display on a GUI acquired images and stock images in a grayscale format for images from perspectives that have yet to be acquired. FIGS. **7A-7F** illustrate an exemplary GUI for acquiring images and/or information and receiving commands from a user. In this embodiment, the software, preferably stored on a non-transitory computer-readable medium, such as a memory device associated with computer **103**, implements the commands and acquires the images for use in generating an inventory or showroom experience by manipulating front-end image acquisition component **102**. In another embodiment, computer **103** functions as a repository for images acquired by front-end image acquisition component **102** so that a user may manipulate the images in various ways. In another embodiment, computer **103** functions as an intermediary between front-end image acquisition component **102** and server **104**. Further embodiments of computer **103** are disclosed herein and in U.S. patent application Ser. No. 14/088,939, incorporated herein by reference above.

(27) The server **104** of FIG. **1** hosts (e.g., stores and/or distributes) the acquired images for use by consumers or sellers via the Internet. Server **104** receives acquired images from computer **103** and stores them on computer-readable media. In an embodiment, server **104** inventories, compresses, stitches, or otherwise processes the received images in addition to, or in the absence of, such actions by computer **103** or front-end image acquisition component **102**. Further embodiments of server **104** are disclosed herein and in U.S. patents application Ser. Nos. 16/385,329, 15/012,033, 14/575,260, and 14/088,939, incorporated herein by reference above.

(28) Referring further to FIG. **1**, database **105** is associated with server **104** for organizing stored images and content. Database **105** may reside on server **104** or on an external computing device that is connected to server **104** via a communications channel. Database **105** stores the acquired images along with metadata or other corresponding information relating to the images and/or the objects themselves. For example, database **105** can store information corresponding to imaged vehicles including, by way of example and not limitation, vehicle stock number, vehicle identification number (VIN), vehicle color, vehicle make, vehicle model, vehicle type, objective vehicle condition information, and the like. In one embodiment, the information contained in database **105** is used with a website template stored on server **104** or an external server for showcasing the objects to consumers via a virtual showroom. Database **105** is capable of using various standards, such as SQL, ODBC, and JDBC, for example. Exemplary database management

systems (DBMS) include MySQL, Microsoft SQL Server, Oracle, and SAP. Database **105** and the information it contains is accessible via the Internet, such as through the use of a web browser or an API. In an exemplary embodiment, database **105** is accessible to end-user console portals **106** and/or an administrator console portal **107**, as disclosed in U.S. patents application Ser. Nos. 14/088,939, 14/575,260, and 14/574,638, incorporated herein by reference above. Further embodiments of database **105** are disclosed herein and in U.S. patents application Ser. Nos. 14/088,939, 14/575,260, and 14/574,638, incorporated herein by reference above.

(29) In an embodiment, database **105** is configured such that the acquired images and data stored in the database can be mined for patterns, analogous values, and the like. In an exemplary embodiment, computing devices implementing machine learning algorithms and/or other data analysis techniques (e.g., image processing, pattern recognition, natural language processing, artificial intelligence, etc.) analyze the images and/or data stored in database **105**, as disclosed in U.S. patent application Ser. No. 15/945,828, incorporated herein by reference above. By way of example only, the computing devices can be programmed to identify defects or damaged components in an imaged vehicle.

(30) The consumer-side portal **106** shown in FIG. **1** provides a means for users, such as consumers, to access the acquired images, content, and information stored in database **105**. In one embodiment, consumer-side portal **106** is a computing device, such as a personal computer, a mobile device, a tablet computing device, and the like. Further embodiments of consumer-side portal **106** are disclosed herein and in U.S. patent application Ser. Nos. 16/385,329, 15/012,033, 14/575,260, and 14/088,939, incorporated herein by reference above. It is to be understood that the users of consumer-side portal **106** need not be purchasers of goods or real estate. Instead, such users include inspectors, inventory/property managers, maintenance personnel, etc.

(31) As shown in FIG. **1**, administrator console portal **107** provides a means for managing information stored in database **105** and/or on server **104**. In one embodiment, the information contained in database **105** is used with administrator console portal **107** stored on server **104** or an external server for managing an inventory of objects or acquired images. Such an inventory management application advantageously allows for easily tracking the progress of an object among the various stages of automated studio **100** (e.g., image acquisition, image processing, and inventory creation). In one embodiment, a seller can execute an administrator console portal **107** application remotely (e.g., via a web application) to provide an automated ticketing process for tracking object progress. Administrator console portal **107** also provides metrics and reporting data and tracks objects, customers, and process information. In addition, administrator console portal **107** provides access to detailed logs and reporting to troubleshoot object-processing errors. The metadata and other corresponding information relating to the images and/or the objects permit a user to determine the location of image and video files. Further embodiments of administrator console portal **107** are disclosed herein and in U.S. patents application Ser. Nos. 16/385,329, 15/012,033, 14/575,260, and 14/088,939, incorporated by reference above.

(32) In an embodiment, automated studio **100** includes server **104**, database **105**, consumer-side portal access **106**, and administrator console portal **107** in addition to front-end image acquisition component **102** and computer **103**.

(33) In addition to the embodiments described above, embodiments of the present disclosure may comprise a special purpose computer including a variety of computer hardware, as described in greater detail below.

(34) Embodiments within the scope of the present disclosure also include computer-readable media for carrying or having computer-executable instructions or data structures stored thereon. Such computer-readable media can be any available media that can be accessed by a special purpose computer. By way of example, and not limitation, computer-readable storage media include both volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures,

program modules or other data. Computer storage media are non-transitory and include, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable ROM (EEPROM), compact-disk ROM (CD-ROM), digital versatile disks (DVD), or other optical disk storage, solid state drives (SSDs), magnetic cassettes, magnetic tape, magnetic disk storage, or other magnetic storage devices, or any other medium that can be used to carry or store desired program code means in the form of computer-executable instructions or data structures and that can be accessed by a general purpose or special purpose computer. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a computer, the computer properly views the connection as a computer-readable medium. Thus, any such connection is properly termed a computer-readable medium. Combinations of the above should also be included within the scope of computer-readable media. Computer-executable instructions comprise, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing device to perform a certain function or group of functions.

(35) The following discussion is intended to provide a brief, general description of a suitable computing environment in which aspects of the disclosure may be implemented. Although not required, aspects of the disclosure will be described in the general context of computer-executable instructions, such as program modules, being executed by computers in network environments. Generally, program modules include routines, programs, objects, components, data structures, etc. that perform particular tasks or implement particular abstract data types. Computer-executable instructions, associated data structures, and program modules represent examples of the program code means for executing steps of the methods disclosed herein. The particular sequence of such executable instructions or associated data structures represent examples of the program code means for executing steps of the methods disclosed herein. The particular sequence of such executable instructions or associated data structures represent examples of corresponding acts for implementing the functions described in such steps.

(36) Those skilled in the art will appreciate that aspects of the disclosure may be practiced in network computing environments with many types of computer system configurations, including personal computers, hand-held devices, multi-processor systems, microprocessor-based or programmable consumer electronics, network PCs, minicomputers, mainframe computers, and the like. Aspects of the disclosure may also be practiced in distributed computing environments where tasks are performed by local and remote processing devices that are linked (either by hardwire links, wireless links, or by a combination of hardwired or wireless links) through a communications network. In a distributed computing environment, program modules may be located in both local and remote memory storage devices.

(37) An exemplary system for implementing aspects of the disclosure includes a special purpose computing device in the form of a conventional computer, including a processing unit, a system memory, and a system bus that couples various system components including the system memory to the processing unit. The system bus may be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, and a local bus using any of a variety of bus architectures. The system memory includes nonvolatile and volatile memory types. A basic input/output system (BIOS), containing the basic routines that help transfer information between elements within the computer, such as during start-up, may be stored in ROM. Further, the computer may include any device (e.g., computer, laptop, tablet, PDA, cell phone, mobile phone, a smart television, and the like) that is capable of receiving or transmitting an IP address wirelessly to or from the internet.

(38) The computer may also include a magnetic hard disk drive for reading from and writing to a removable magnetic disk, a magnetic disk drive for reading from or writing to a removable magnetic disk, and an optical disk drive for reading from or writing to removable optical disk such as a CD-ROM or other optical media. The magnetic hard disk drive, magnetic disk drive, and

optical disk drive are connected to the system bus by a hard disk drive interface, a magnetic disk drive-interface, and an optical drive interface, respectively. The drives and their associated computer-readable media provide nonvolatile storage of computer-executable instructions, data structures, program modules, and other data for the computer. Although the exemplary environment described herein employs a magnetic hard disk, a removable magnetic disk, and a removable optical disk, other types of computer readable media for strong data can be used, including magnetic cassettes, flash memory cards, digital video disks, Bernoulli cartridges, RAMs, ROMs, SSDs, and the like.

(39) Communication media typically embody computer readable instructions, data structures, program modules or other data in a modulated data signal such as a carrier wave or other transport mechanism and includes nay information discovery media.

(40) One or more aspects of the disclosure may be embodied in computer-executable instructions (i.e., software), routines, or functions stored in system memory or nonvolatile memory as application programs, program modules, and/or program data. The software may alternatively be stored remotely, such as on a remote computer with remote application programs. Generally, program modules include routines, programs, objects, components, data structures, etc. that perform particular tasks or implement particular abstract data types when executed by a processor in a computer or other device. The computer executable instructions may be stored on one or more tangible, non-transitory computer readable media (e.g., hard disk, optical disk, removable storage media, solid state memory, RAM, etc.) and executed by one or more processors or other devices. As will be appreciated by one of skill in the art, the functionality of the program modules may be combined or distributed as desired in various embodiments. In addition, the functionality may be embodied in whole or in part in firmware or hardware equivalents such as integrated circuits, application specific integrated circuits, field programmable gate arrays (FPGA), and the like.

(41) The computer may operate in a networked environment using logical connections to one or more remote computers. The remote computers may each be another personal computer, a tablet, a PDA, a server, a router, a network PC, a peer device, or other common network node, and typically include many or all of the elements described above relative to the computer. The logical connections include a local area network (LAN) and a wide area network (WAN) that are presented here by way of example and not limitation. Such networking environments are commonplace in office-wide or enterprise-wide computer networks, intranets, and the Internet.

(42) When used in a LAN networking environment, the computer is connected to the local network through a network interface or adapter. When used in a WAN networking environment, the computer may include a modem, a wireless link, or other means for establishing communications over the WAN, such as the Internet. The modem, which may be internal or external, is connected to the system bus via the serial port interface. In a networked environment, program modules depicted relative to the computer, or portions thereof, may be stored in the remote memory storage device. It will be appreciated that the network connections shown are exemplary and other means of establishing communications over WAN may be used.

(43) Preferably, computer-executable instructions are stored in a memory, such as the hard disk drive, and executed by the computer. Advantageously, the computer processor has the capability to perform all operations (e.g., execute computer-executable instructions) in real time.

(44) The order of execution or performance of the operations in the embodiments illustrated and described herein is not essential, unless otherwise specified. That is, the operations may be performed in any order, unless otherwise specified, and embodiments may include additional or fewer operations than those disclosed herein. For example, it is contemplated that executing or performing a particular operation before, contemporaneously with, or after another operation is within the scope of aspects of the disclosure.

(45) Embodiments may be implemented with computer-executable instructions. The computer-executable instructions may be organized into one or more computer-executable components or

modules. Aspects of the disclosure may be implemented with any number and organization of such components or modules. For example, aspects of the disclosure are not limited to the specific computer-executable instructions or the specific components or modules illustrated in the figures described herein. Other embodiments may include different computer-executable instructions or components having more or less functionality than illustrated and described herein.

(46) FIG. 12A illustrates a detailed criteria data structure **1200**, for criterion identifiers and corresponding values that are associated with an item of interest, in accordance with an embodiment of the invention. In one embodiment, detailed criteria data structure **1200** is adapted to provide data representing an item of interest (e.g., a vehicle) in an objective manner.

(47) The detailed criteria data structure **1200** comprises a first group of criterion identifier fields **1202**, which include data that represents various criterion identifiers by which the item of interest may be identified or evaluated. For example, the criterion identifier fields **1202** may include data representing criterion identifiers that ascertain the item of interest, such as a VIN, make, model, and the like of a vehicle. Further, criterion identifier fields **1202** may include data representing evaluating criteria such as a percentage of an item of interest that is damaged and dimensions of damaged areas. It is to be understood by one skilled in the art that the criterion identifier fields **1202** may also include data representing evaluating criteria that are subjective, such as condition categories and opinions on the value of the item of interest.

(48) Referring further to FIG. 12A, the detailed criteria data structure **1200** also has a second group of criterion value fields **1204** including data that represents values associated with a corresponding criterion identifier field **1202**. For example, the criterion identifier field **1202-C** includes data representing a criterion identifier of “color” and corresponding criterion value field **1204-C** includes data representing a value of “red.”

(49) In one embodiment, detailed criteria data structure **1200** is stored in database **105** on server **104**. In a further embodiment, criterion identifier fields **1202** and criterion value fields **1204** include data that is presented to a user via various components of a comprehensive condition report.

(50) One embodiment of detailed criteria data structure **1200** provides a benefit of reduced network bandwidth usage. This benefit may be provided by detailed objective information of detailed criteria data structure **1200** and allows finite communications network resources to be used to carry out other tasks. Additionally, detailed criteria data structure **1200** may reduce the usage of computing resources of server **104** while still providing users with desired information about a desired item or item of interest. Moreover, increased user efficiency provided by detailed criteria data structure **1200** results in reduced usage of stored electrical energy in a battery powering a display device in end-user console portals **106** on which data from detailed criteria data structure **1200** is displayed. In another embodiment, detailed objective information provided by detailed criteria data structure **1200** about an item of interest allows consumers to realize advantages of both the traditional goods or real estate purchasing approach and the virtual showroom approach. Further, detailed criteria data structure **1200** can improve the efficiency of automated search processes involving database **105**, as discussed in greater detail below with respect to FIG. 12C.

(51) FIG. 12B illustrates a detailed criteria component **1210**, which is adapted for determining values that are represented by data included in criterion identifier fields **1202** and criterion value fields **1204**. In the embodiment illustrated by FIG. 12B, the detailed criteria component **1210** is presented in tangible form as a computer-executable module on server **104**. However, it is to be understood by one skilled in the art that detailed criteria component **1210** may also be implemented as a digital or analog electronic circuit. Detailed criteria component **1210** comprises sub-components, including a criterion identifier component **1212**, a criterion value component **1214**, a value determination component **1216**, and a memory **1218**. Also included in this embodiment is a network interface **1220** of server **104**.

(52) The network server interface **1220** is communicatively connected to front-end information acquisition components **102** and database **105** via communications networks, such as the Internet.

The network interface **1220** is also communicatively connected to the memory **1228**, the criterion identifier component **1212**, the criterion value component **1214**, and the value determination component **1216**. In this manner, network interface **1220** is also communicatively connected to detailed criteria component **1210**. These communicative connections between network interface **1220** and other components included within server **104** may be accomplished via electrical interconnects. Memory **1218** is also electrically connected to criterion identifier component **1212**, criterion value component **1214**, and value determination component **1216**. Further, criterion identifier component **1212**, criterion value component **1214**, and value determination component **1216** are each electrically connected to one another.

(53) The network interface **1220** is adapted for facilitating data communication between front-end information acquisition components **102**, database **105**, and detailed criteria component **1210** and its various sub-components. Memory **1218** is adapted for storing data utilized and/or generated by criterion identifier component **1212**, criterion value component **1214**, and value determination component **1216**.

(54) Criterion identifier component **1212** is adapted for providing data representing a current criterion identifier, such as criterion identifier field **1202-F** that corresponds to a largest scratch length value of a scratch on a vehicle of interest. Value determination component **1216** is adapted for determining a value corresponding to the current criterion identifier provided by criterion identifier component **1212**. In one embodiment, further explained below with respect to FIG. **12C**, value determination component **1216** utilizes data provided by criterion identifier component **1212** and an image obtained from front-end information acquisition components **102** via network interface **1220** to determine the value corresponding to the current criterion identifier provided by criterion identifier component **1212**.

(55) Criterion value component **1214** is adapted for providing data representing the determined current criterion value. In one embodiment, criterion value component **1214** provides data representing the value in criterion value field **1204-F** that corresponds to a value of “6” for the largest scratch length value of a scratch on the vehicle of interest. The data provided by criterion value component **1214** may be received from value determination component **1216**. The data provided by criterion value component **1214** may also be received from front-end information acquisition components **102** via either network interface **1220** or memory **1218**.

(56) In an embodiment, detailed criteria component **1210** provides advantages including increased user efficiency, reduced network bandwidth usage, and reduced computing resource utilization due to detailed information provided by detailed criteria component **1210**.

(57) Referring to FIG. **12C**, a flow chart illustrates a method of determining values that are represented by data included in criterion value fields **1204**. The method includes an image acquisition step **1250**, a feature detection step **1252**, an image segmentation step **1254**, a feature extraction step **1256**, a feature identification and classification step **1258**, a feature measurement step **1260**, a feature dimension output step **1262**, an area determination step **1264**, a feature percentage step **1266**, and a feature percentage output step **1268**. In one embodiment, the method steps are embodied in computer-executable instructions executing on server **104**. In another embodiment, the method steps may also be implemented by digital or analog electronic circuitry.

(58) In operation according to one embodiment, an image is obtained at image acquisition step **1250** and stored in a computer-readable format. For example, an image is acquired with front-end information acquisition components **102** and transmitted to server **104**. Upon completion of the image acquisition step **1250**, feature detection step **1252** is performed. In one embodiment, feature detection step **1252** comprises examining each pixel of the acquired image to determine whether a feature is present at a given pixel. Exemplary image features include, but are not limited to, edges, corners (e.g., interest points), regions of interest (e.g., blobs), and ridges. Feature detection step **1252** may utilize feature detection methods such as Canny, smallest univalue segment assimilating nucleus (SUSAN) and features from accelerated segment test (FAST) embodied as computer-

executable instructions, for example. Alternatively, feature detection step **1252** may be facilitated by methods that involve accessing photographs and/or associated vehicle information from other vehicles previously acquired and stored in database **105**.

(59) In one embodiment, after feature detection step **1252** is completed, image segmentation step **1254** is performed. During image segmentation step **1254**, the obtained image is transformed by a partitioning into multiple sets of pixels in order to locate objects and boundaries (e.g., lines, curves) in the obtained image. It is to be understood by one skilled in the art that while image segmentation step **1254** may be omitted and the method may proceed from feature detection step **1252** directly to feature extraction step **1256**.

(60) Referring further to FIG. **12C**, the feature extraction step **1256** transforms the obtained image into a set of features, such that the set contains relevant information from the obtained image. For example, feature extraction step **1256** may transform an obtained image of a vehicle of interest into a set of scratches on the vehicle. In an embodiment, feature extraction step **1256** transforms the obtained image into a set of features with minimal resources of server **104** and memory **1218**. In one embodiment, feature extraction step **1256** comprises general dimensionality reduction techniques, such as principal component analysis and latent semantic analysis, for example. In another embodiment, feature extraction step **1256** utilizes a set of application-dependent features.

(61) After feature extraction step **1256**, the method illustrated by FIG. **12C** proceeds to the feature identification and classification step **1258**, in which one feature from the set of features is identified and classified as being of a certain type. In an embodiment, feature identification and classification step **1258** selects one feature from the set of features produced by feature extraction step **1256** and associates a unique identifier with that feature. The unique identifier given to the selected feature allows that feature to be tracked among more than one image and to be distinguished among other features. The unique identifier can further be tracked with respect to analogous features belonging to another vehicle or vehicles, whether such information is already stored in database **105** or will be collected and/or stored at a later time. When tracked in this manner, the unique identifier given to each selected feature can facilitate machine learning algorithms and/or other data analysis techniques implemented by a processor, which can be used to train the processor for identifying specific features on the vehicle's exterior. In another embodiment, feature identification and classification step **1258** associates a selected feature with a type of features. For example, a feature that corresponds to a scratch on a vehicle of interest may be classified as type "scratch." The classification allows multiple features of the same type to be associated with each other for categorization. In one embodiment, feature identification and classification step **1258** utilizes data provided by criterion identifier component **1212** to provide the classification of the feature.

(62) Still referring to the method illustrated by FIG. **12C**, the dimensions of the identified feature are determined by feature measurement step **1260**. In an embodiment, feature measurement step **1260** determines the dimensions of the identified feature in terms of pixels. In a further embodiment, feature measurement step **1260** utilizes a predetermined value that is a function of the distance of the imaging device from the imaged object in order to determine the dimensions of the identified feature in real-world units (e.g., inches, centimeters, etc.). In yet another embodiment, feature measurement step **1260** utilizes a known real-world dimension of an object in the same image as the identified feature in order to determine the dimensions of the identified feature in real-world units. The determined dimensions of the identified feature are feature dimensions **1262**. In an embodiment, feature dimensions **1262** are provided to criterion value component **1216**.

(63) The embodiment of FIG. **12C** further determines the total area of a region of interest, which includes the identified feature, at area determination step **1264**. In an exemplary embodiment, area determination step **1264** uses edges detected in feature detection step **1252** to determine four edges that comprise the perimeter of a rectangular area of interest. The dimensions of the four edges are determined by feature measurement step **1260**. Then, using the dimensions, area determination step **1264** uses the equation that the area of a rectangle is equal to the width of the rectangle multiplied

by the length of the rectangle (i.e., $A=W \times L$) to calculate the area of the rectangular area of interest. It is to be understood by one skilled in the art that the total area of regions having other shapes may also be determined. For example, circular areas (i.e., $A=\pi \cdot \text{sup.}^2$) and irregularly shaped areas e.g., using a definite integral) may also be determined. In one embodiment, area determination step **1264** determines the area of the region of interest in the unit of pixels. In another embodiment, area determination step **1264** utilizes a predetermined value that is a function of the distance of the imaging device from the imaged object in order to determine the dimensions of the identified feature in real-world units (e.g., inches, centimeters, etc.). In yet another embodiment, area determination step **1264** utilizes a known real-world dimension of an object in the same image as the identified feature in order to determine the dimensions of the identified feature in real-world units.

(64) The feature percentage step **1266** then determines the ratio of the identified feature to the area of the region of interest. In an exemplary embodiment, feature percentage step **1266** determines the area of an item of interest that is scratched and divides that value by the total area of the item of interest to calculate the ratio of the item of interest that is scratched. The determined ratio of the identified feature to the area of the region of interest is feature percentage **1268**. In an embodiment, feature percentage **1268** is provided to criterion value component **1216**.

(65) FIG. **13A** illustrates a desired criteria data structure **1300** for criteria and corresponding values that are associated with a desired item of interest according to an embodiment of the invention. The desired criteria data structure **1300** has a first group of desired criterion identifier fields **1302** comprising data that represents criterion identifiers by which the desired item of interest may be identified or evaluated. In one exemplary embodiment, desired criterion identifier fields **1302** include data representing identifying criteria such as a make, model, color, and the like of a vehicle. Additionally, desired criterion identifier fields **1302** may include data representing evaluating criteria such as a desired percentage level of damage to an item of interest and acceptable dimensions of damaged areas. Moreover, desired criterion identifier fields **1302** may also include data representing evaluating criteria that are subjective, such as acceptable condition categories (e.g., mint, average, fine, poor) and desired opinions on the value of the item of interest (e.g., CONSUMERS DIGEST BEST BUY™).

(66) Referring again to FIG. **13A**, desired criteria data structure **1300** also includes a second group of desired criterion value fields **1304**, each including data that represents values associated with a corresponding desired criterion identifier field **1302**. For example, desired criterion identifier field **1302-G** includes data representing a criterion of a maximum scratch width, in inches, acceptable to a user and corresponding desired criterion value field **1304-G** includes data representing a value of "0.25." Desired criterion value fields **1304** may also contain data representing acceptable values expressed in other ways. For example, "less than" and "greater than" values, such as "less than 15" of desired criterion value field **1304-H**, the "any" of desired criterion value field **1304-A** indicating that a user does not have a preference for the corresponding criterion **1302-A**, and the range "0-5" of desired criterion value field **1304-E**.

(67) In one embodiment, desired criteria data structure **1300** functions as a wish list or favorites list of desired or acceptable criteria and corresponding values for a desired item of interest. In one embodiment, desired criteria data structure **1300** is stored in database **105** on server **104**. In a further embodiment, desired criterion identifier fields **1302** and desired criterion value fields **1304** include data that are presented to a user via various components of a comprehensive condition report. In another embodiment, desired criteria data structure **1300** functions as a model for a machine-learning algorithm for identifying features belonging to particular classes of vehicles.

(68) One embodiment of desired criteria data structure **1300** provides a benefit of reduced network bandwidth usage. This benefit may be provided by desired criterion values provided by desired criteria data structure **1300** and allows finite communications network resources to be used to carry out other tasks. Further, desired criteria data structure **1300** may reduce the usage of computing

resources on server **104** while still providing users with the ability to locate a desired item or item of interest. Additionally, increased user efficiency provided by desired criteria data structure **1300** results in reduced usage of stored electrical energy in a battery powering a display device in end-user console portals **106**. In another embodiment, desired criteria data structure **1300** allows consumers to realize advantages of both the traditional goods or real estate purchasing approach and the virtual showroom approach. Further, detailed criteria data structure **1300** can improve the efficiency of automated search processes involving database **105**, as discussed in greater detail below with respect to FIG. **13C**.

(69) FIG. **13B** illustrates a desired criteria component **1310** that, in one embodiment, is adapted for determining when an actual criterion value matches a desired criterion value. In a further embodiment, desired criteria component **1310** is adapted for determining when an amount of matched criterion values satisfies an acceptable threshold level. The desired criteria component **1310** includes a desired criterion identifier component **1312**, a desired criterion value component **1314**, an actual criterion value component **1316**, a match component **1318**, a threshold component **1320**, and a memory **1322**. In the embodiment of FIG. **13B**, desired criteria component **130** is included within server **104**, in addition to a network interface **1324**.

(70) The network interface **1324** is communicatively connected to front-end information acquisition components **102** and database **105** via communications networks, such as the Internet. Network interface **1324** is also communicatively connected to the desired criterion identifier component **1312**, the desired criterion value component **1314**, the actual criterion value component **1316**, the match component **1318**, the threshold component **1320**, and the memory **1322**. By virtue of network interface **1324** being communicatively connected to the sub-components of desired criteria component **1310**, one skilled in the art will understand that network interface **1324** can be said to also be communicatively connected to desired criteria component **1310**. Memory **1322** is electrically connected to desired criterion identifier component **1312**, desired criterion value component **1314**, actual criterion value component **1316**, match component **1318**, and threshold component **1320**. Desired criterion identifier component **1312** is electrically connected to desired criterion value component **1314**, actual criterion value component **1316**, and match component **1318**. Desired criterion value component **1314** and actual criterion value component **1316** are each electrically connected to match component **1318**. Moreover, threshold component **1320** is electrically connected to match component **1318**.

(71) Referring further to FIG. **13B**, the network interface **1324** is adapted for facilitating data communication between front-end information acquisition components **102**, database **105**, and desired criteria component **1310** and its sub-components. Memory **1322** is adapted for storing data utilized and/or generated by desired criterion identifier component **1312**, desired criterion value component **1314**, actual criterion value component **1316**, match component **1318**, and threshold component **1320**. In an embodiment, memory **1322** is a memory component on server **104** that may be utilized by additional components of server **104**. In another embodiment, memory **1322** is a memory component utilized solely by desired criteria component **1310**.

(72) Desired criterion identifier component **1312** is adapted for providing data representing a current desired criterion identifier, such as desired criterion identifier field **1302-H** corresponding to a maximum desired percentage of hail damage to a desired vehicle. In one embodiment, desired criterion identifier component **1312** receives the current desired criterion identifier from database **105** via network interface **1324**. In a further embodiment, desired criterion identifier component **1312** receives the current desired criterion identifier from memory **1322**.

(73) Desired criterion value component **1314** of FIG. **13B** is adapted for providing data representing a desired criterion value corresponding to the current desired criterion identifier provided by desired criterion identifier component **1312**. In an embodiment, desired criterion value component **1314** provides data representing desired criterion value field **1304H**, which corresponds to a value of the maximum desired percentage of hail damage to the desired vehicle. The desired

criterion value **1314** may be acquired from database **105** and/or may be generated by a processor at least in part based on previously acquired vehicle information stored in database **105**.

(74) Actual criterion value component **1316** is adapted for providing data representing an actual criterion value corresponding to the current desired criterion identifier provided by desired criterion identifier component **1312**. In one embodiment, actual criterion value component **1316** provides data representing criterion value field **1204-H**, which corresponds to a value of the percentage of hail damage to a vehicle of interest.

(75) Still referring to FIG. **13B**, match component **1318** is adapted for determining when the actual criterion value provided by actual criterion value component **1316** satisfies the desired criterion value provided by desired criterion value component **1314**. In an embodiment, the data provided by desired criterion identifier component **1312**, desired criterion value component **1314**, and actual criterion value component **1316** are received by match component **1318**. Match component **1318** then compares the actual criterion value and the desired criterion value, as further described below with respect to FIG. **13C**. In an embodiment in which the actual criterion value satisfies the desired criterion value, match component **1318** provides the desired criterion identifier and a match indicator to memory **1322**, threshold component **1320**, and/or network interface **1324**. In an embodiment in which the actual criterion value fails to satisfy the desired criterion value, match component **1318** provides the desired criterion identifier and a no-match indicator to memory **1322**, threshold component **1320**, and/or network interface **1324**.

(76) Threshold component **1320** is adapted for determining when a number of match indicators provided by match component **1318** and/or memory **1322** satisfies a threshold value provided by memory **1322** and/or network interface **1324**. In one embodiment, data provided by match component **1318** is received by threshold component **1320**. In a further embodiment, data stored in memory **1322** is retrieved by threshold component **1320**. Threshold component **1320** compares a percent of match indicators to a threshold value, as further described below with respect to FIG. **13D**.

(77) In one embodiment, a threshold value is provided by an end-user console portal **106** and stored in database **105** or memory **1322**. In another embodiment, a threshold value is produced by an executed algorithm that compares previously acquired vehicle information, such as photographs and/or related metadata stored in database **105**, and then generates an idealized value based on specified parameters, such as (by way of example only) a value representing a predefined percentile of all vehicles compared by the algorithm. The threshold value indicates an amount of deviation from a desired item that will be considered acceptable. For example, desired criteria data structure **1300** may represent criterion values of a desired vehicle, such that satisfaction of every criterion value by a vehicle of interest would be deemed a complete match. However, a deviation from a complete match, such as a vehicle of interest that satisfies ninety percent of the criterion values, may be acceptable for a particular need. In this instance, the vehicle of interest will be deemed a matching vehicle although it is not a complete match.

(78) In an embodiment in which the percent of match indicators satisfies the threshold value, threshold component **1320** provides a match indication. For example, the match indication may be provided by threshold component **1320** to memory **1322**, database **105** via network interface **1324**, and/or end-user console portals **106** via network interface **1324**. In an embodiment in which the percent of match indicators does not satisfy the threshold value, threshold component **1320** does not provide a match indication.

(79) In an embodiment, desired criteria component **1310** provides advantages including increased user efficiency, reduced network bandwidth usage, and reduced computing resource utilization due to data provided by match component **1318** and/or threshold component **1320**.

(80) Referring next to FIG. **13C**, a flow chart illustrates a method of determining a match between a desired criterion value and an actual criterion value. In one embodiment, the illustrated method is embodied in computer-executable instructions stored on server **104** and executed by match

component **1318**. The method includes a read criterion identifier step **1330**, a read actual criterion value step **1332**, a read desired criterion value step **1334**, a determination step **1336**, a store match indicator step **1338**, and a store no-match indicator step **1340**.

(81) During the read criterion identifier step **1330**, data representing a desired criterion identifier is accessed or received. In an exemplary embodiment, match component **1318** receives the data representing the desired criterion identifier provided by desired criterion identifier component **1312**. At the read actual criterion value step **1332**, data representing an actual criterion value is accessed or received. For example, match component **1318** receives the data representing the actual criterion value provided by actual criterion value component **1316**. The read desired criterion value step **1334** results in data representing a desired criterion value being accessed or received. In an embodiment, match component **1318** receives the data representing the desired criterion value provided by desired criterion value component **1314**.

(82) At determination step **1336** of FIG. **13C**, it is ascertained whether the actual criterion value satisfies the desired criterion value. For example, match component **1318** may compare an actual value of “10” (e.g., the percentage of hail damage to a vehicle of interest in criterion value field **1204-H**) to a desired value of “less than 15” (e.g., the maximum desired percentage of hail damage to a desired vehicle in desired criterion value field **1304-H**). In this example, the actual value “10” satisfies the desired value of “less than 15,” and thus the vehicle of interest matches the desired vehicle with respect to the maximum percentage of hail damage criterion identifier.” Such a satisfaction of the desired criterion value results in the method proceeding to step **1338**, in which the criterion identifier and a match indicator are stored in memory **1322** and/or database **105**. The match indicator specifies that the desired criterion value has been satisfied by the actual criterion value for the corresponding criterion identifier.

(83) In the event determination step **1336** ascertains that the actual criterion value does not satisfy the desired criterion value, the method proceeds to step **1340**. At step **1340**, the criterion identifier and a no-match indicator are stored in memory **1322** and/or database **105**. The no-match indicator specifies that the actual criterion value has failed to satisfy the desired criterion value for the corresponding criterion identifier.

(84) Referring to FIG. **13D**, a flow chart illustrates a method of determining when an item of interest falls with a desired match threshold value. As explained above with respect to FIG. **13B**, the threshold value indicates an amount of deviation from a desired item that will be considered acceptable. In one embodiment, the method of FIG. **13D** is embodied in computer-executable instructions stored on server **104** and executed by threshold component **1320**. The method includes a match indicator amount step **1350**, a total combined amount step **1352**, a percent determination step **1354**, a threshold value step **1356**, a satisfaction determination step **1358**, a match indication step **1360**, and an end process step **1362**.

(85) At the match indicator amount step **1350**, a determination is made of an amount of the number of match indicators for a particular item of interest. In one embodiment, threshold component **1320** accesses match indicators stored in memory **1322** and determines a total number of match indicators associated with an item of interest.

(86) During the total combined amount step **1352** of FIG. **13D**, a determination is made of a total combined number of match indicators and no-match indicators associated with the item of interest. For example, threshold component **1320** may access the match indicators and no-match indicators stored in memory **1322** and then add the amount of each together.

(87) At percent determination step **1354**, a determination is made of the ratio of match indicators to the total combined number of match indicators and no-match indicators. In an embodiment, the ratio indicates an amount that an item of interest deviates from a desired item. During threshold value step **1356**, a threshold value is accessed by threshold component **1320**. In one embodiment, threshold component **1320** accesses the threshold value from memory **1322**. Threshold component **1320** may also access the threshold value from database **105** or receive the threshold value end-user

console portals **106** via network interface **1324**.

(88) Still referring to FIG. **13D**, a satisfaction determination step **1358** ascertains whether the percentage determined at percent determination step **1354** satisfies the threshold value from threshold value step **1356**. For example, a percentage of 95 percent may be determined during percent determination step **1354** and a threshold value of 90 percent may be accessed at threshold value step **1356**. Thus, the item of interest deviates from a desired item by 5 percent (i.e., the difference between 100 and 95) and the amount of deviation from the desired item that is acceptable is 10 percent (i.e., the difference between 100 and 90). In this example, the percentage of 95 percent satisfies the threshold value of 90 percent. Such a satisfaction of the threshold value results in the method proceeding to step **1360**, in which a match indication is provided. In an embodiment, a match indication is provided as an alert on a GUI of a device of end-user console portals **106**. A match indication may also be an e-mail, a text message, or the like. In another embodiment, the match indication is provided to database **105** via network interface **1324**. In the event satisfaction determination step **1358** ascertains that the percentage does not satisfy the threshold value, the method proceeds to step **1362** where the process ends.

(89) FIG. **2A** illustrates a mobile image acquisition apparatus **200**, which in one embodiment serves as the front-end image acquisition component **102** of FIG. **1**. The mobile image acquisition apparatus **200** includes a backpack **202**, camera **210**, light source **220**, support and positioning cord **230**, umbilical cable **240**, electrical power source **250**, wireless communications transceiver **260**, computing device **270**, and computing device data cable **280**. Mobile image acquisition apparatus **200** may also be used in conjunction with a support member **290**.

(90) In an embodiment, mobile image acquisition apparatus **200** is adapted for acquiring images of the interior of a vessel **292** that has an opening **294**. The opening **294** allows ingress and egress of portions of mobile image acquisition apparatus **200** relative to the interior of the vessel **292**. In the embodiment of FIG. **2A**, vessel **292** is an oil tanker railcar and opening **294** is a manway.

(91) Referring further to FIG. **2A**, the light source **220** is physically connected to the camera **210**. The support and positioning cord **230** is physically connected to camera **210** and support member **290**. The umbilical cable **240** is electrically connected to light source **220** and camera **210** as well as to the electrical power source **250**, wireless communications transceiver **260**, and computing device data cable **280**. Computing device data cable is electrically connected to umbilical cable **240**, wireless communications transceiver **260**, and computing device **270**. Camera **210** is in electrical communication with wireless communications transceiver **260** via umbilical cable **240**. Further, camera **210** is in electrical communication with computing device **270** via umbilical cable **240** and computing device data cable **280**. Computing device **270** is in electrical communication with wireless communications transceiver **260** via computing device data cable **280**. Electrical power source **250** is electrically connected to wireless communications transceiver **260**.

(92) The support member **290** of FIG. **2A** is adapted for being positioned at a point above opening **294** of vessel **292** such that support and positioning cord **230** and other portions of mobile imaging apparatus **200** are also positioned above opening **294** of vessel **292**. Support member **290** is also adapted for being moved into positions closer to opening **294** or farther away from opening **294** and/or for being moved in a plane that is parallel to the plane of opening **294**. In the embodiment of FIG. **2**, support member **290** is the hand of a human user.

(93) As shown in FIG. **2A**, support member **290** is a structure of rigid members assembled together to create a frame-like structure. In an alternative embodiment, shown by FIG. **2B**, support member **290** is included with a structure of legs **282** that are rigid members assembled together to create a tripod **284** that may straddle opening **294** of vessel **292**. In the embodiment of FIG. **2B**, support member **290** is connected to rigid members **282** at the apex of the tripod **284**. Support member **290** may alternatively be attached to another machine that facilitates acquiring images of the interior of vessel **292** while vessel **292** is in motion or transit. In another embodiment, support member **290** may be an unmanned aerial vehicle.

(94) Referring to FIG. 2A, vessel **292** may be any receptacle that contains materials during transit for a variety of transportation methods. For example, vessel **292** may be a railcar, a shipping container, a semi-trailer, or an intermodal freight container. In another aspect, vessel **292** may be any container that holds hazardous materials during transportation of the hazardous materials. Opening **294** may be any gap in the body of vessel **292**. For example, opening **294** may be a door opening, a valve, an outlet, a vent, and the like.

(95) As shown in FIG. 2A, support and positioning tether **230** is adapted for supporting camera **210**, light source **220**, and portions of umbilical cable **240**. Support and positioning tether **230** is also adapted for transferring a corresponding manipulation of support member **290** to camera **210**, light source **220**, and portions of umbilical cable **240**. For example, support and positioning tether **230** may be a nylon-covered cable, a steel cable, or the like. Support and positioning tether **230** may be wound around a spool or a reel in order to facilitate easier lengthening and shortening. In the embodiment of FIG. 2B, support and positioning tether **230** is connected to tripod **284**. Support and positioning tether **230** may be used to lower tripod **284** through opening **294** of vessel **292**. In another embodiment, support and positioning tether **230** may be used to retrieve tripod **284**, and attached components, from vessel **292**.

(96) In an embodiment, support and positioning tether **230** may be comprised of a rigid material, such as one or more pieces of aluminum welded or bolted together. In another embodiment, support and positioning tether **230** comprised of a rigid material may telescope. Advantageously, telescoping functionality allows mobile image acquisition apparatus **200** to remain compact during storage or transit while providing the ability to position camera **210** and light source **220** a greater distance from support member **290**. In an embodiment, support and positioning tether **230** is adapted for providing an extension to position camera **210** and light source **220** inside vessel **292** while keeping other components of mobile image acquisition apparatus **200** outside vessel **292**. Support and positioning tether **230** may also be of such a thickness that it does not readily appear in stitched-together images acquired from inside vessel **292**.

(97) The camera **210** is adapted for acquiring images of the interior portion of a wall of vessel **292** and/or other objects within the cavity of vessel **292**. In the embodiment of FIG. 2A, two cameras **210** having fisheye lenses are mounted back-to-back and physically connected to the end of support and positioning cord **230** opposite support member **290**. Each fisheye camera **210** allows essentially a hemisphere of viewing and preferably converts the distorted hemispherical image into a conventional rectilinear projection. As used herein, images include still photographs (e.g., snapshots), moving photographs (e.g., videos or movies), panoramas, stereoscopic photographs, infrared images, or any combination thereof. In addition to acquiring images of the entire interior portion of vessel **292**, images of only a portion of an interior portion of vessel **292** may also be acquired. For example, areas of the interior that are particularly noteworthy, such as fissures, scratches, dents, hail damage, welded joints, and the like. Further, camera **210** may also acquire images of the exterior of vessel **292**. For example, images of the top of the exterior or the bottom of the exterior, such as the undercarriage, axles, and wheels may also be acquired by camera **210**. In the embodiment of FIG. 2B, camera **210** may be connected to support member **290** at a connection point **284**.

(98) In an embodiment, camera **210** is configured to convert a distorted hemispherical image into other projections, such as cylindrical, spherical, or other specialized projections. Camera **210** may be a programmable digital single-lens reflex (DSLR) camera that provides pan, tilt, and zoom capabilities. Suitable DSLR cameras are available from Nikon Corporation and Canon, Inc. Camera **210** may also be an Internet Protocol (IP) camera adapted for connecting to a telecommunications network, for example one utilizing the IP communications protocol. Suitable IP cameras are available from GeoVision, Inc. Arecont Vision provides suitable IP video cameras and associated software. Camera **210** may also be adapted for directly coupling to computing device **270** via a communications channel employing serial and/or parallel communications

methods. Camera **210** may also utilize other lens types. In addition, various filters, such as polarizing filters, may be employed on camera **210** for improving image quality. Advantageously, software on camera **210**, computing device **270**, an external computing device, or any combination thereof, is capable of automatically stitching the acquired images together to form a continuous image that accurately depicts a 360-degree view of the interior of vessel **292**. In another embodiment, camera **210** comprises a power source and does not utilize power source **250**, described below.

(99) In the embodiment of FIG. 2A, light source **220** is adapted for providing sufficient lighting conditions inside the cavity of vessel **292** for acquisition of images by camera **210**. The light source **220** is physically mounted on cameras **210** and is comprised of, for example, a plurality of light-emitting diodes (LEDs). In another embodiment, light source **220** is independent of cameras **210** and, for example, suspended from its own support structure through opening **294** or an alternate opening in vessel **292**. In the embodiment of FIG. 2B, light source **220** is connected to support member **290** and suspended from tripod **284** such that it is adapted for entering vessel **292**. In another embodiment, more than one light source **220** is utilized, with one light source **220** physically mounted on cameras **210** and another light source **220** suspended from its own support structure.

(100) In an embodiment, light source **220** is a single LED. Light source **220** may also be an incandescent light bulb, a compact fluorescent light bulb, and the like. In another embodiment light source **220** also emits electromagnetic waves outside the visible portion of the electromagnetic spectrum.

(101) The umbilical cable **240** of FIG. 2A is adapted for transmitting electrical power from electrical power source **250** to camera **210** and light source **220**. Umbilical cable **240** is also adapted for transmitting electrical signals between camera **210**, light source **220**, wireless communications transceiver **260**, computing device data cable **280**, and computing device **270**. In an embodiment, umbilical cable **240** transmits acquired images from camera **210** to wireless communications transceiver **260** and computing device **270** and also transmits control signals from computing device **270** to camera **210** and/or light source **220**. In the embodiment of FIG. 2A, umbilical cable **240** is comprised of an Ethernet cable and an electrical power cable that are enclosed in a protective sheath and having an exposed connection location at a point along the length of umbilical cable **240**. Advantageously, these exposed connections allow camera **210**, light source **220**, support and positioning cord **230**, and a portion of umbilical cable **240** to be disconnected from the remaining portion of umbilical cable **240** to facilitate user access, storage, and portability of mobile image acquisition apparatus **200**. The protective sheath facilitates cable management and organization and also provides protection against edges of opening **294** and/or other portions or contents of vessel **292**.

(102) In the embodiment of FIG. 2B, umbilical cable **240** is connected to support member **290** and suspended from tripod **284** such that a connector is available near connection point **284**. Umbilical cable **240** may also be comprised of other cables, including Universal Serial Bus (USB), Ethernet, and the like. Umbilical cable **240** may also be used without a protective sheath. Umbilical cable **240** may also be used with a protective sheath that does not have an exposed connection location.

(103) Electrical power source **250** is adapted for providing electrical energy to power the various components of mobile image acquisition apparatus **200**, including camera **210**, light source **220**, and wireless communications transceiver **260**. In the embodiment of FIG. 2A, electrical power source **250** is a battery and is located in backpack **202** worn by a user of mobile image acquisition apparatus **200**. In the embodiment of FIG. 2B, electrical power source **250** is connected to one leg **282** of tripod **284** and is adapted for providing electrical energy to light source **220**. In an embodiment, electrical power source **250** is adapted such that electrical energy is provided to various components of image acquisition apparatus **200** while requiring only one connection to an external power source for charging. In one embodiment, electrical power source **250** is a solar

power system.

(104) Referring further to FIG. 2A, wireless communications transceiver **260** is adapted for transmitting acquired images to a server or database via a communications medium, such as the Internet. For example, wireless communications transceiver **260** may transmit acquired images to server **104** and/or database **105** of FIG. 1. Additionally, wireless communications transceiver **260** is adapted for receiving control signals via a communications medium, such as the Internet. The control signals provide information regarding the operation and manipulation of camera **210** and/or light source **220**. In the embodiment of FIG. 2, wireless communications transceiver **260** is the 341U cellular network modem available from Netgear, Inc. and is located within backpack **202**. Advantageously, wireless communications transceiver **260** allows mobile image acquisition apparatus **200** to receive and send information without the need for wires and to operate in a variety of locations. Another benefit of wireless communications transceiver **260** is the ability to immediately upload acquired images to a server or other remote storage means across a communications network (e.g., the Internet) for processing. This ability provides savings in the cost, labor, and time required for image acquisition and processing.

(105) It is to be understood that wireless communications transceiver **260** is configured to operate according to, for example, the IEEE 802.11 (Wi-Fi) standard and/or the IEEE 802.15.1

(Bluetooth™) standard. Embodiments of wireless communications transceiver **260** also include a cellular network modem or a computing device having wireless communications capabilities.

(106) Computing device **270** is adapted for executing computer-readable instructions embodied in software, adapted for displaying, on a graphical user interface (GUI), a systematic process for acquiring images of the interior of vessel **292**, and adapted for receiving commands from a user to acquire the images. FIGS. 7A-7F illustrate an exemplary GUI for acquiring images and/or information and receiving commands from a user. In the embodiment of FIG. 2, computing device **270** is, for example, a laptop computer having its own electrical power source and braced near the user's body by computing device support strap **272** that is worn around the user's neck and physically connected to computing device **270**. Computing device support strap **272** is adapted for supporting the weight of computing device **270**. In the embodiment of FIG. 2A, computing device support strap **272** is comprised of polypropylene webbing and utilizes strap adjusters to facilitate lengthening or shortening of the strap. Advantageously, support strap **272** allows computing device **270** to be located in an easily accessible location for the user without the need for a separate support surface and also allows the user to keep both hands free.

(107) In one embodiment, computing device **270** is a processor. In another embodiment, computing device **270** is an integrated circuit containing one or more processing units, a memory, an arithmetic logic unit, and/or a control unit.

(108) Computing device data cable **280** is adapted for transmitting electrical signals between umbilical cable **240**, wireless communications transceiver **260**, and computing device **270**. In the embodiment of FIG. 2, computing device data cable **280** is an Ethernet cable that is connected to a portion of umbilical cable **240** and computing device **270**. Advantageously, computing device data cable **280** allows computing device **270** to be disconnected from umbilical cable **240** to facilitate user access, storage, and portability of mobile image acquisition apparatus **200**.

(109) Computing device data cable **280** is preferably embodied by a communications channel employing serial and/or parallel communications methods. And computing device **270** is configured such that the transmission of data between umbilical cable **240**, wireless communications transceiver **260**, and computing device **270** can be accomplished via wireless communications methods, such as the IEEE 802.11 (Wi-Fi) standard and the IEEE 802.15.1 (Bluetooth™) standard.

(110) Referring to FIG. 2B, tripod **284** includes three legs **282**. In an embodiment, legs **282** may be comprised of a rigid material, such as one or more pieces of aluminum affixed together. In another embodiment, legs **282** comprised of a rigid material may telescope. Advantageously, telescoping

functionality allows tripod **284** to remain compact during storage or transit, while still providing the ability to position camera **210** and light source **220** inside an opening **294** of vessel **292**. In an embodiment, tripod **284** is collapsed and affixed to backpack **202** during transport or storage. In another embodiment, tripod **284** is adapted for providing an extension to position camera **210** and light source **220** inside vessel **292** while keeping other components of mobile image acquisition apparatus **200** outside vessel **292**.

(111) During use according to one embodiment, a user manually moves mobile image acquisition apparatus **200** or a portion thereof to position it relative to vessel **292**. In another embodiment, support member **290** is attached to a machine or vehicle that self-locomotes or support member **290** is itself a machine or vehicle that self-locomotes. Advantageously, self-locomotive capabilities allow the mobile image acquisition apparatus to be moved without the need for manual movement by a human user and also allows movement commands to be transmitted from a remote computing device or stored on computing device **270**.

(112) Mobile image acquisition apparatus **200** may also transmit acquired images to a server or database. Mobile image acquisition apparatus **200** may also include a barcode scanner or a radio-frequency identification (RFID) reader that communicates with other portions of mobile image acquisition apparatus **200** via wireless communications methods, umbilical cable **240**, or an additional communications cable. Advantageously, a barcode scanner or an RFID reader allows for the collection of information identifying vessel **292** and the integration of that information with acquired images of vessel **292**. In an embodiment, mobile image acquisition apparatus **200** is adapted for acquiring images of a vessel that is the subject of an accident or disaster investigation or rescue mission.

(113) In operation of an embodiment, support member **290** is positioned at a point above opening **294** of vessel **292** such that support and positioning cord **230** and portions of mobile imaging apparatus **200** are also positioned above opening **294** of vessel **292**. Then support member **290** is moved to a position closer to opening **294** such that portions of support and positioning cord **230** and portions of mobile imaging apparatus **200** enter a cavity of vessel **292** through opening **294**. In an embodiment, camera **210**, light source **220**, and portions of umbilical cable **240** are the portions of mobile imaging apparatus **200** that enter the cavity of vessel **292** through opening **294**.

(114) Once camera **210** and light source **220** are positioned inside the cavity of vessel **292**, light source **220** illuminates the interior portion of a wall of vessel **292**. Camera **210** acquires images of the interior portion of a wall of vessel **292** and/or other objects within the cavity of vessel **292**. Support member **290** and/or support and positioning cord **230** are manipulated in order to focus or point camera **210** and/or light source **220** at various locations of the interior portion of vessel **292**. Functions specific to camera **210** (e.g., zoom and pan) are also utilized via computing device **270** to further focus camera **210** at the various locations. In addition to these functions, camera **210** is also otherwise manipulated from commands on computing device **270** that are communicated to camera **210** via computing device data cable **280** and/or umbilical cable **240**. Moreover, various filters for camera **210** are also implemented from commands on computing device **270** that are communicated to the camera **210** via computing device data cable **280** and/or umbilical cable **240**. In addition, light source **220** is controlled from commands on computing device **270** that are communicated to light source **220** via computing device data cable **280** and/or umbilical cable **240**.

(115) The image data collected by camera **210** is then transmitted to wireless communications transceiver **260** and computing device **270** via umbilical cable **240** and computing device data cable **280**. The image data is then wirelessly transmitted to an external receiver via wireless communications transceiver **260**. This wireless transmission to the external receiver allows the acquired images to be uploaded to a remote server or storage device without the need to wait until the user finishes the image collection process. Beneficially, the immediate uploading of images allows the images to be processed sooner than conventional techniques. The image processing techniques described in U.S. patents application Ser. Nos. 15/012,033, 14/574,638, and 14/088,939

may be used for this purpose. The image data may also be displayed via a GUI on computing device **270**.

(116) FIGS. **3A** and **3C** illustrate a turntable **301** and at least one camera **302**, which in one embodiment serve as the front-end image acquisition component **102** of FIG. **1**. Turntable **301** comprises a rotatable platform adapted for receiving and supporting, for example, a vehicle and rotating the vehicle 360 degrees as well as portions and iterations thereof. According to aspects of the invention, a relay provides an interface between turntable **301** and computer **103** or a communications network transmitting control information. Upon receiving an appropriate command or control signal from computer **103** or communications network, turntable **301** rotates the desired angular distance at the desired angular velocity, thus rotating the vehicle a uniform distance and at a uniform velocity. Suitable turntables include those described in U.S. Pat. Nos. 6,817,300 and 7,631,602. FIG. **3B** illustrates turntable **301** supporting a vehicle and located within an enclosure having a ceiling **303** and walls **304**, further described herein and in U.S. patent application Ser. No. 14/088,939, incorporated herein by reference above. Contemplated vehicles include, by way of example and not limitation, automobiles, boats, watercraft, recreational vehicles, motorcycles, all-terrain vehicles, trailers, aircraft, and the like.

(117) The cameras **302** are adapted for acquiring images of a vehicle supported by turntable **301**, preferably as the vehicle rotates via turntable **301**. As used herein, images include still photographs (e.g., snap shots), moving photographs (e.g., videos or movies), panoramics, stereoscopic photographs, infrared images, or any combination thereof. Preferably, cameras **302** are configured for acquiring images of an entire vehicle or, if desired, images of a portion of a vehicle. For example, images may be taken of the VIN, text identifying the model of the vehicle, or any areas of the vehicle that are particularly noteworthy, such as scratches, dents, hail damage, add-ons, tire treads, and the like. It is also contemplated that a salesperson can provide a video demonstration of the vehicle's features and options to be integrated into the final virtual showroom. The video "sales pitch" can be made while the vehicle is rotating via turntable **301** or stationary.

(118) Referring to FIG. **3C**, cameras **302** preferably include one or more programmable DSLR cameras that provide pan, tilt, and zoom capabilities. Suitable DSLR cameras are available from Nikon Corp. and Canon, Inc. Cameras **302** may also be Internet protocol (IP) cameras adapted for connecting to a telecommunications network, for example one utilizing the Internet Protocol communications protocol, such as the Internet. Suitable IP cameras are available from GeoVision, Inc. Arecont Vision provides suitable IP video cameras and associated software. It is also contemplated that cameras **302** are capable of directly coupling to a computing device via a relay or a communications channel employing serial and/or parallel communications methods. The cameras **302** preferably utilize a wide-angle lens, including, by way of example and not limitation, a fisheye lens. The present embodiment also contemplates the use of one or more handheld cameras. In addition, various filters, such as polarizing filters, may be employed on cameras **302** for improving image quality.

(119) In an embodiment illustrated by FIG. **3C**, at least one of cameras **302** is located at a fixed position outside the circumference of turntable **301** and is pointed generally toward the center of turntable **301**. The fixed camera **302** is preferably affixed at a height between five feet and six and one-half feet above turntable **301** to provide a view from the perspective of an average consumer if the consumer were viewing the vehicle in-person. Advantageously, affixing the camera **302** at a fixed height while acquiring images provides the same perspective for each image so that a user is provided with a constant perspective when viewing the acquired images. Additionally, images having the same perspective provide a consistent and seamless view when the acquired images are stitched together to provide a better user experience. As turntable **301** rotates the vehicle, fixed camera **302** acquires images of the vehicle's exterior. Advantageously, software executed by cameras **302**, computer **103**, server **104**, or any combination thereof, is capable of automatically stitching the acquired images together to form a continuous image that accurately depicts a 360-

degree view of the exterior of the vehicle. Further embodiments of the cameras **302** are disclosed herein and in U.S. patent application Ser. Nos. 15/012,033, 14/574,638, and 14/088,939, incorporated herein by reference above.

(120) In another embodiment, cameras **302** utilize a processor to facilitate receiving and transmitting electronic control data and acquired image data. For example, cameras **302** may be affixed to a tower structure that houses computer **103**. In an embodiment, cameras **302** may utilize computer **103**, which displays on a GUI a systematic process for acquiring images of an object and acquired images. FIGS. 7A-7F illustrate an exemplary GUI for acquiring images and/or information and receiving commands from a user. Computer **103** may also display on a GUI stock images in a grayscale format for images of the object from perspectives that have yet to be acquired. In another embodiment, cameras **302** utilize an integrated circuit containing one or more processing units, a memory, an arithmetic logic unit, and/or a control unit to facilitate receiving and transmitting electronic control data and acquired image data.

(121) FIGS. 4A-B illustrate an exemplary imaging studio **400**, which in one embodiment serves as the front-end image acquisition component **102** of FIG. 1. FIG. 4A provides a side view perspective, while FIG. 4B provides a top view perspective of the imaging studio **400**. The imaging studio **400** includes a stationary camera **410**, a movable camera **420**, subject alignment patterns **430A-C**, camera alignment patterns **432A-F**, an alignment facilitator **434**, a structure **440**, and a support surface **442**. Beneficially, the imaging studio **400** is capable of installation in a pre-existing structure. For example, the imaging studio **400** may be installed in an automotive service bay. This capability provides the advantage of utilizing a service bay as an imaging space when there is no demand to use the bay to service automobiles. As a result, the service bay owner can increase the uses to which the space can be put, optimize resource usage, and increase overall profit. Additionally, the use of a pre-existing structure reduces the initial capital investment required for the imaging studio **400** and allows for easy removal and/or resale of various components of the imaging studio **400**.

(122) The stationary camera **410** in the illustrated embodiment is physically connected to structure **440**. Movable camera **420**, subject alignment patterns **430A-C**, and camera alignment patterns **432A-F** are located upon support surface **442**. The subject being imaged (e.g., a vehicle) is also supported by support surface **442**.

(123) According to aspects of the invention, stationary camera **410** is adapted for acquiring images of the subject. Stationary camera **410** is physically attached at a fixed location to structure **440** via bolts, screws, welding, or the like. Stationary camera **410** preferably includes one or more DSLR cameras that provide pan, tilt, and zoom capabilities. Suitable DSLR cameras are available from Nikon Corp. and Canon, Inc. Stationary camera **410** may also include one or more Internet protocol (IP) cameras adapted for connecting to a telecommunications network, for example one utilizing the Internet Protocol communications protocol, such as the Internet. Suitable IP cameras are available from GeoVision, Inc. Arecont Vision provides suitable IP video cameras and associated software. In another embodiment, stationary camera **410** is communicatively connected to computer **103**, receives electronic command signals from computer **103**, and transmits acquired image data to computer **103**. In another embodiment, stationary camera **410** is adapted for directly coupling to computer **103** via a relay or communications channel employing serial and/or parallel communications methods. In yet another embodiment, stationary camera **410** transmits acquired image data to server **104** and/or database **105**. In addition, various filters, such as polarizing filters, may be employed on stationary camera **410** for improving image quality.

(124) Referring further to FIGS. 4A and 4B, movable camera **420** is adapted for acquiring images of the exterior and/or interior of the subject. In one embodiment, movable camera **420** is imaging apparatus **500** (see FIGS. 5A-5C) described herein. In another embodiment, movable camera **420** is imaging apparatus **550** (see FIG. 5D) described herein. In another embodiment, movable camera **420** is one or more handheld cameras. Preferably, movable camera **420** transmits acquired image

data to computer **103**, server **104**, and/or database **105**. In one embodiment, movable camera **420** and stationary camera **410** acquire images of the same subject substantially simultaneously. In another embodiment, movable camera **420** and stationary camera **410** acquire images of the same subject at different times. For example, stationary camera **410** may acquire images of the subject and after that image acquisition process is complete, movable camera **420** may acquire images of the same subject.

(125) The subject alignment patterns **430A-C**, as illustrated by FIG. **4B**, are adapted for providing a means to consistently position subjects for imaging with respect to camera alignment patterns **432A-F**. The embodiment of FIG. **4B** is preferably utilized to acquire images of a vehicle. In this embodiment, subject alignment pattern **430A** is designed to be used for vehicles with a small footprint or wheelbase (e.g., compact car), subject alignment pattern **430B** is designed to be used for vehicles with a medium footprint or wheelbase (e.g., sedan), and subject alignment pattern **430C** is designed to be used for vehicles with a large footprint or wheelbase (e.g., truck). Each subject alignment pattern **430A-C** contains a short portion that runs parallel to the axles of the vehicle and represents where the front wheels of the vehicle should be located. Each subject alignment pattern **430A-C** also contains a longer portion that runs perpendicular to the axles of the vehicle and represents where the edge of the footprint of the vehicle should be located. In one embodiment, each subject alignment pattern **430A-C** is a different color to allow a human user to easily distinguish the patterns and determine which subject alignment pattern **430A-C** to utilize for a particular vehicle. For example, subject alignment pattern **430A** may be blue, subject alignment pattern **430B** may be green, and subject alignment pattern **430C** may be red. In a further embodiment, the different color of each subject alignment pattern **430A-C** may correspond to a matching colored marking device located upon the vehicle. In this way, a determination can be made as to which subject alignment pattern **430A-C** should be utilized with a particular vehicle so that when a driver enters the vehicle to drive it into position, he or she can immediately know which subject alignment pattern **430A-C** to utilize. One skilled in the art will recognize that subject alignment patterns **430A-C** may be comprised of markings of various shapes and in various numbers to accommodate various subjects to be imaged.

(126) The camera alignment patterns **432A-F**, as illustrated by FIG. **4B**, are adapted for providing a means to consistently position movable camera **420** with respect to a subject being imaged and subject alignment patterns **430A-C**. The embodiment of FIG. **4B** is preferably utilized to acquire images of a vehicle. In this embodiment, each camera alignment pattern **432A-F** comprises two markings that indicate where the front and sides of movable camera **420** should be located. In operation of this embodiment, movable camera **420** is initially positioned at camera alignment pattern **432A**. After images are acquired by movable camera **420** at camera alignment pattern **432A**, movable camera **420** is repositioned at camera alignment pattern **432B** where movable camera **420** acquires additional images. This operation is repeatable by repositioning movable camera **420** at each of the remaining camera alignment patterns **432C-F**. The movable camera may be repositioned at each of camera alignment patterns **432A-F** in order, out of order, or by skipping some of the camera alignment patterns **432A-F**. In another embodiment, camera alignment patterns **432A-F** are footprints that indicate where a human user using a handheld camera should stand to acquire images of the subject being imaged. Advantageously, subject alignment patterns **430A-C** and camera alignment patterns **432A-F** allow a user to quickly position subjects to be imaged and movable camera **420** so that images can be acquired from a consistent perspective. Camera alignment patterns **432A-F** may be comprised of more or less alignment patterns depending upon the subject to be imaged.

(127) The alignment facilitator **434** is adapted to aid positioning of the subject being imaged. In one embodiment, alignment facilitator **434** is a mirror used in conjunction with subject alignment patterns **430**. For example, a driver of a vehicle that is the subject being imaged may use alignment facilitator **434** embodied as a mirror for determining when the center of one wheel of the vehicle,

and thus the entire vehicle, is in a suitable position with respect to subject alignment patterns **430**. In other embodiments, alignment facilitator **434** may be an electronic sensor.

(128) The structure **440**, as illustrated by FIG. **4A**, is adapted for providing physical support for stationary camera **410**. In this embodiment, structure **440** provides a means for stationary camera **410** to acquire images of the subject from a consistent location above the imaged subject. In one embodiment, structure **440** is the ceiling of a pre-existing automotive service bay.

(129) Support surface **442** is adapted for receiving and supporting the subject to be imaged and for supporting movable camera **420**, subject alignment patterns **430A-C**, and camera alignment patterns **432A-F**. In one embodiment, support surface **442** is the floor of a pre-existing automotive service bay. In another embodiment, support surface **442** is a turntable, such as turntable **301**, described herein and in U.S. patent application Ser. No. 14/088,939, incorporated herein by reference above.

(130) In another embodiment, stationary camera **410** and/or movable camera **420** utilize a processor to facilitate receiving and transmitting electronic control data and acquired image data. For example, stationary camera **410** may be affixed to a tower structure that houses computer **103**. In an embodiment, stationary camera **410** and movable camera **420** may utilize computer **103**, which displays on a GUI a systematic process for acquiring images of an object and acquired images (e.g., ceiling shots, undercarriage shots, etc.). FIGS. **7A-7F** illustrate an exemplary GUI for acquiring images and/or information and receiving commands from a user. Computer **103** may also display on a GUI stock images in a grayscale format for images of the object from perspectives that have yet to be acquired. In another embodiment, stationary camera **410** and/or movable camera **420** utilize an integrated circuit containing one or more processing units, a memory, an arithmetic logic unit, and/or a control unit to facilitate receiving and transmitting electronic control data and acquired image data.

(131) In one embodiment, a method for acquiring images for rendering a virtual vehicle showroom comprises positioning a vehicle at a first predetermined location upon a support surface, positioning a mobile imaging apparatus at a second predetermined location relative to the vehicle, acquiring a plurality of images of the vehicle with the mobile imaging apparatus at the second predetermined location, positioning the mobile imaging apparatus at a third predetermined location relative to the vehicle, and acquiring a plurality of images of the vehicle with the mobile imaging apparatus at the third predetermined location. In another embodiment, the method further comprises transforming the acquired images into a stitched 360-degree panoramic image.

(132) FIG. **5A** illustrates an imaging apparatus **500**, which in one embodiment serves as the front-end image acquisition component **102** of FIG. **1**. In the illustrated embodiment, the imaging apparatus **500** includes a mobile cart **505**, a support arm **501**, a linear motion member **504**, one or more cameras **502**, one or more light sources **503**, a barcode scanner **508**, one or more wireless communications transceivers **506**, and an electric power source **507**.

(133) In an embodiment, imaging apparatus **500** is adapted for positioning cameras **502** and light sources **503** (e.g., LEDs) inside a vehicle or the like to acquire images of the vehicle's interior and for transmitting the images to server **104** and/or database **105**. In another embodiment, imaging apparatus **500** is adapted for positioning cameras **502** and light sources **503** at various locations around the perimeter of the vehicle to acquire images of the vehicle's exterior and for transmitting the images to server **104** and/or database **105**. Advantageously, an embodiment of imaging apparatus **500** allows acquisition of images of the interior of the particular vehicle being added to the online inventory, rather than using stock images or a representative model. This advantage allows consumers to view, for example, a particular vehicle that is for sale and provides them with more information for their purchasing decision.

(134) Referring further to FIG. **5A**, mobile cart **505** is adapted for providing a support structure upon which to affix various components of imaging apparatus **500** and a means with which to position imaging apparatus **500** relative to the vehicle being imaged. Mobile cart **505** is comprised

of rigid members assembled together to create a frame-like structure. By way of example and not limitation, mobile cart **505** may be comprised of pieces of aluminum welded or bolted together. Mobile cart **505** may have a padding material, such as foam, affixed to the rigid material to provide a buffer that prevents the rigid material from scratching anything, such as a vehicle, during use of the cart **505**. In one embodiment, the frame-like structure of mobile cart **505** is enclosed in a protective housing to make imaging apparatus **500** waterproof and shockproof. Other components of imaging apparatus **500** may be permanently or temporarily affixed to the frame structure of mobile cart **505**, which provides portability and self-containment for imaging apparatus **500**.

(135) During use according to one embodiment, a user manually moves mobile cart **505** to position imaging apparatus **500** relative to the vehicle or other subject to be imaged. For example, the user moves cart **505** to position cameras **502** inside a vehicle having a view of substantially the entire vehicle interior. In another embodiment, mobile cart **505** self-locomotes via a motor, which drives a device that enables movement of imaging apparatus **500** across a surface. By way of example and not limitation, an electric motor drives wheels, skid-steer tracks, or the like to move imaging apparatus **500** across a surface to position imaging apparatus **500** relative to the vehicle. In another embodiment, mobile cart **505** remains stationary while support arm **501** is positioned relative to the vehicle. Advantageously, a user familiar with the system can acquire the images needed to generate a 360-degree view of the vehicle's interior in a short amount of time (e.g., less than two minutes).

(136) In an embodiment, support arm **501** is adapted for providing an extension to position cameras **502** and light sources **503** inside the vehicle while keeping the other components of imaging apparatus **500** outside the vehicle. In another embodiment, support arm **501** is adapted for providing an extension to position cameras **502** and light sources **503** at various points around the exterior of the vehicle, such as above or below the vehicle. Support arm **501** is comprised of rigid members assembled together. Support arm **501** may be comprised of pieces of the same material as cart **505** or may be comprised of different materials. Support arm **501** is of such a thickness that it does not readily appear in stitched-together images acquired, for example, from inside the vehicle. An exemplary thickness of support arm **501** is less than two inches (e.g., 1.875 inches).

(137) According to aspects of the invention, support arm **501** has a portion that extends in a vertical direction and a portion that extends in a horizontal direction and is affixed at some point to cart **505**. By way of example and not limitation, support arm **501** may extend vertically upward from cart **505** and then make a 90-degree bend and extend horizontally away from cart **505**. In an embodiment, support arm **501** is one single, unitary piece. In another embodiment, support arm **501** is adapted for moving in a vertical direction by a linear motion member **504**. Linear motion member **504** moves support arm **501** via a motor drive, such as a stepper motor drive.

Advantageously, linear motion member **504** allows support arm **501** to be moved vertically so cameras **502** and light sources **503** can be positioned through an open window of the vehicle at varying heights. In another embodiment, support arm **501** telescopes in a vertical direction and/or a horizontal direction. Advantageously, the telescoping capability of support arm **501** allows imaging apparatus **500** to remain compact while providing the ability to position cameras **502** and light sources **503** a greater distance from mobile cart **505**. In one embodiment, support arm **501** allows cameras **502** and light sources **503** to be positioned above a vehicle, such that images can be acquired of the roof of the vehicle. In another embodiment, support arm **501** allows cameras **502** and light sources **503** to be positioned under a vehicle, such that cameras **502** can acquire images of the undercarriage of the vehicle. In this embodiment, support arm **501** may extend vertically downward from cart **505** and then make a bend that is substantially 90 degrees and extend horizontally away from cart **505**.

(138) The one or more cameras **502** are adapted for acquiring images of the interior of the vehicle. In the embodiment of FIGS. 5A and 5B, two cameras **502** having fisheye lenses are mounted back-to-back on the end of the horizontal portion of support arm **501** farthest from the vertical portion. Each fisheye camera **502** allows essentially a hemisphere of viewing and preferably converts the

distorted hemispherical image into a conventional rectilinear projection. It is contemplated that other projections may be used, such as cylindrical, spherical, or other specialized projections. Advantageously, software executed by cameras **502**, computer **103**, server **104**, or any combination thereof, automatically stitches the images together to form a continuous image that accurately depicts the interior of the vehicle. Cameras **502** preferably include one or more programmable DSLR cameras that provide pan, tilt, and zoom capabilities. Suitable DSLR cameras are available from Nikon Corp. and Canon, Inc. Cameras **502** may also be one or more Internet protocol (IP) cameras adapted for connecting to a telecommunications network, for example one utilizing the Internet Protocol communications protocol, such as the Internet. Suitable IP cameras are available from GeoVision, Inc. Arecont Vision provides suitable IP video cameras and associated software. It is also contemplated that cameras **502** are capable of directly coupling to a computing device via a relay or a communications channel employing serial and/or parallel communications methods. Cameras **502** preferably utilize a wide-angle lens, including, by way of example and not limitation, a fisheye lens. The present embodiment also contemplates the use of one or more handheld cameras. In addition, various filters, such as polarizing filters, may be employed on cameras **502** for improving image quality.

(139) The one or more light sources **503** are adapted for providing sufficient lighting conditions inside the vehicle for acquisition of the images. In the embodiment of FIGS. 5A and 5B, light sources **503** are mounted on support arm **501** adjacent to cameras **502**. Preferably, light sources **503** are comprised of light-emitting diodes.

(140) Referring to FIGS. 5A and 5C, barcode scanner **508** is adapted for scanning the VIN of the vehicle. Advantageously, barcode scanner **508** is handheld and allows a user to collect the VIN without the need to manually enter the VIN into computer **103** or database **105**. In an embodiment, acquiring the VIN will initiate the image acquisition process.

(141) The one or more wireless communications transceivers **506** are adapted for transmitting acquired images to server **104** and/or database **105** via a communications medium, such as the Internet. Additionally, wireless communications transceivers **506** are adapted for receiving control signals generated by computer **103** or another computing device via a communications medium, such as the Internet. The control signals provide information regarding the relative position of cart **505** with respect to the vehicle, regarding the vertical movement of support arm **501**, and regarding the operation and manipulation of cameras **502** and light sources **503**. Wireless communications transceivers **506** are affixed to cart **505** and may be any transceiver capable of receiving and/or transmitting communications signals. By way of example and not limitation, wireless communications transceivers **506** may operate according to the IEEE 802.11 (WiFi) standard, the IEEE 802.15.1 (Bluetooth™) standard, may be a cellular network modem, or may be a laptop computer having wireless communications capabilities. A suitable cellular network modem is the 341U available from Netgear, Inc. Advantageously, wireless communications transceivers **506** allow imaging apparatus **500** to receive and send information without the need for wires and to operate in a variety of locations.

(142) The electric power source **507** is preferably adapted for providing electrical energy to power the various components of the imaging apparatus, including the electric motor of cart **505**, the motor of linear motion member **504**, cameras **502**, light sources **503**, and wireless communications transceivers **506**. Electric power source **507** is affixed to cart **505**. Preferably, electric power source **507** is a battery that can be recharged or replaced, such as, by way of example and not limitation, an automotive battery. In another embodiment, electric power source **507** is a photovoltaic cell.

(143) In another embodiment, various components of imaging apparatus **500**, including one or more cameras **502**, utilize a processor to facilitate receiving and transmitting electronic control data and acquired image data. In another embodiment, various components of imaging apparatus **500**, including one or more cameras **502**, utilize an integrated circuit containing one or more processing units, a memory, an arithmetic logic unit, and/or a control unit to facilitate receiving and

transmitting electronic control data and acquired image data.

(144) In one embodiment, imaging apparatus **500** is positioned at a predefined location. The positioning may be accomplished by a user manually moving imaging apparatus **500** or by imaging apparatus **500** self-locomoting. The predefined location is one of a plurality of predefined locations surrounding the footprint of the vehicle. In one embodiment, the predefined locations comprise markings on a floor surface. In one embodiment, imaging apparatus **500** is programmed via computer-executable instructions executing on computer **103** to self-locomote around the perimeter of a vehicle being imaged and acquire images of the vehicle.

(145) In another embodiment, once in the predefined location, computer **103** provides a user a list of preset image selections that correspond to a portion of the vehicle being imaged. Once the user selects a desired preset image selection, computer **103** transmits control data to cameras **502** for focusing on the corresponding portion of the vehicle and acquiring an image. For example, a preset image selection may be “front driver-side headlight,” which corresponds to the front driver-side headlight on the vehicle. The user selects a graphical button representing “front driver-side headlight” on a graphical user interface (GUI) of computer **103** and computer **103** sends a command to cameras **502** to focus on the front driver-side headlight of the vehicle. A preview image of the front driver-side headlight is shown to the user via the GUI of computer **103**. An exemplary and non-limiting list of preset image selections for a vehicle includes headlight, taillight, tire tread, vehicle side, grill, emblem/logo, mirror, door handle, wheels, tailgate, doors, engine, and trunk. For selections such as engine and trunk, a door or enclosure may need to be manually opened before an image is acquired. In an alternative embodiment, once the user is satisfied with the preview image, the user selects another graphical button on the GUI of computer **103** that results in computer **103** sending a command to cameras **502** to acquire an image.

(146) FIG. 5D illustrates imaging apparatus **550**, which is another embodiment of imaging apparatus **500** and in one embodiment serves as the front-end image acquisition component **102** of FIG. 1. One skilled in the art will appreciate that various components of imaging apparatus **500** and imaging apparatus **550** may be interchanged for each other or combined into additional embodiments. Imaging apparatus **550** includes body support component **551**, extension support member **552**, vertical support member **553**, vertical motion member **554**, camera support arm **555**, movable camera and light source **556**, camera housing **557**, stationary camera **558**, warning light **559**, speaker **560**, wireless transceiver **561**, kill switch **562**, access door **563**, wheels **564**, balance support **565**, processor **566**, power source **567**, motor **568**, and sensors **569**. Beneficially, imaging apparatus **550** provides improvements in the field of automated object imaging by providing a means to acquire images of a particular object and upload them to an online inventory or database in a small amount of time and with little or no human involvement.

(147) Referring further to FIG. 5D, the extension support member **552**, camera housing **557**, speaker **560**, wireless transceiver **561**, access door **563**, wheels **564**, balance support **565**, processor **566**, power source **567**, motor **568**, and sensors **569** are each physically connected to body support component **551**. The vertical support member **553**, stationary camera **558**, warning light **559**, and kill switch **562** are each physically connected to extension support member **552**. Vertical motion member **554** is physically connected to vertical support member **553**. Camera support arm **555** is physically connected to vertical motion member **554**. Movable camera and light source **556** is physically connected to camera support arm **555**. Vertical motion member **554**, camera support arm **555**, movable camera and light source **556**, stationary camera **558**, warning light **559**, speaker **560**, wireless transceiver **561**, kill switch **562**, processor **566**, motor **568**, and sensors **569** are each electrically connected to power source **567**. Vertical motion member **554**, camera support arm **555**, movable camera and light source **556**, stationary camera **558**, warning light **559**, speaker **560**, wireless transceiver **561**, kill switch **562**, motor **568**, and sensors **569** are each electrically connected to processor **566**.

(148) In an embodiment, imaging apparatus **550** is adapted for positioning movable camera and

light source **556** inside a vehicle to acquire images of the vehicle's interior and for transmitting the acquired images to server **104** and/or database **105**. In another embodiment, imaging apparatus **550** is adapted for positioning stationary camera **558** at various locations around the perimeter of the vehicle to acquire images of the vehicle's exterior and for transmitting the images to server **104** and/or database **105**. Beneficially, imaging apparatus **550** may acquire images of a vehicle without the need for a human user to manually position imaging apparatus **550** at various locations around the vehicle. Another advantage of imaging apparatus **550** is that it allows images to be acquired of the particular vehicle's interior and exterior so that they can be added to the online inventory instead of using stock images or a representative model. These advantages allow consumers to view the particular vehicle that is for sale and provides them with more information for their purchasing decision.

(149) In FIG. **5D**, body support component **551** is adapted for providing a support structure upon which to affix various components of imaging apparatus **550** and also for providing a means to enclose various components of imaging apparatus **550**. Body support component **551** is comprised of a molded plastic housing that provides both support and enclosure capabilities. Body support component **551** may be comprised of other materials, such as aluminum, steel, composite materials, and the like.

(150) The extension support member **552** is adapted for providing a structural link between vertical support member **553**, vertical motion member **554**, camera support arm **555**, movable camera and light source **556** and body support component **551** that extends horizontally from body support component **551**. In one embodiment, extension support member **552** is affixed to body support component **551** via bolts. Extension support member **552** may also be affixed to body support component **551** via welding or other methods of joining materials. By way of example and not limitation, extension support member **552** may be comprised of pieces of aluminum welded or bolted together and may also be comprised of plastic.

(151) Referring further to FIG. **5D**, vertical support member **553** is adapted for providing a structural link between vertical motion member **554**, camera support arm **555**, and movable camera and light source **556** and extension support member **552**. Vertical support member **553** extends substantially perpendicular to extension support member **552** and is affixed to extension support member **552** and camera housing **557**. In one embodiment, vertical support member **553** is affixed to extension support member **552** and camera housing **557** via bolts. In other embodiments, vertical support member **553** may be affixed to extension support member **552** and camera housing **557** via welding and other methods of joining materials. By way of example and not limitation, vertical support member **553** may be comprised of pieces of aluminum welded or bolted together and may also be comprised of plastic.

(152) In an embodiment, vertical motion member **554** is adapted for providing a means to change the vertical positioning of camera support arm **555** and movable camera and light source **556**. For example, vertical motion member **554** may be an actuator or motor.

(153) The camera support arm **555** is adapted for providing an extension to position movable camera and light source **556** inside the vehicle. In another embodiment, camera support arm **555** is adapted for providing an extension to position movable camera and light source **556** at various points around the exterior of the vehicle, such as above or below the vehicle. Camera support arm **555** is affixed to vertical support member **553** and pivots so that it can extend away horizontally from vertical support member **553** and body support component **551** or so that it can be stowed in a vertical position substantially parallel to vertical support member **553**. In one embodiment, camera support arm **555** is affixed to vertical support member **553** via bolts. In other embodiments, camera support arm **555** is affixed to vertical support member **553** via welding and other methods of joining materials. By way of example and not limitation, camera support arm **555** comprises pieces of aluminum welded or bolted together and/or plastic. Camera support arm **555** is of such a thickness that it does not readily appear in stitched-together images acquired, for example, from

inside the vehicle. An exemplary thickness of camera support arm 555 is 1.875 inches. In another embodiment, camera support arm 555 telescopes, which provides the advantage of allowing it to remain compact while providing the ability to position movable camera and light source 556 a greater distance from body support component 551 and vertical support member 553. In addition, one or more sensors 569 may be affixed to camera support arm 555.

(154) Referring further to FIG. 5D, movable camera and light source 556 is adapted for providing sufficient lighting conditions and acquiring images of the interior of, for example, a vehicle and is comprised in this embodiment of two cameras having fisheye lenses mounted back-to-back at the end of camera support arm 555. Each fisheye camera allows essentially a hemisphere of viewing and preferably converts the distorted hemispherical image into a conventional rectilinear projection. In other embodiments, other projections, such as cylindrical, spherical, or other specialized projections may be used. Advantageously, software executed by movable camera and light source 556, computer 103, server 104, or any combination thereof, automatically stitches the images together to form a continuous image that accurately depicts the interior of the vehicle. Movable camera and light source 556 preferably includes one or more programmable DSLR cameras that provide pan, tilt, and zoom capabilities. Suitable DSLR cameras are available from Nikon Corp. and Canon, Inc. Movable camera and light source 556 may also include one or more Internet protocol (IP) cameras adapted for connecting to a telecommunications network, for example one utilizing the Internet Protocol communications protocol, such as the Internet. Suitable IP cameras are available from GeoVision, Inc. Arecont Vision provides suitable IP video cameras and associated software. It is also contemplated that movable camera and light source 556 is capable of directly coupling to a computing device via a relay or a communications channel employing serial and/or parallel communications methods. The cameras of movable camera and light source 556 preferably utilize a wide-angle lens, including, by way of example and not limitation, a fisheye lens. The present embodiment also contemplates the use of one or more handheld cameras. In addition, various filters, such as polarizing filters, may be employed on the cameras of movable camera and light source 556 for improving image quality. The light source of movable camera and light source 556 is mounted on camera support arm 555 adjacent to the cameras and is preferably comprised of light-emitting diodes. Further, one or more sensors 569 may be affixed on, or integrated into, movable camera and light source 556. For example, sensors 569 may help prevent damage to the subject being imaged by movable camera and light source 556.

(155) The camera housing 557 is adapted for protecting movable camera and light source 556 when it is not in use. Camera housing 557 may be comprised of plastic, aluminum, steel, composite materials, or the like. Preferably, camera housing 557 has a three surfaces to protect movable camera and light source 556, including a bottom portion, a back portion, and an outer side portion. In this configuration, camera housing 557 is open at the top and on the side facing the front of imaging apparatus 550 to allow camera support arm 555 to pivot and place movable camera and light source 556 inside camera housing 557.

(156) In the illustrated embodiment, stationary camera 558 is adapted for acquiring images of the exterior of the vehicle. Stationary camera 558 is affixed to the end of extension support member 552 opposite vertical support member 553. Stationary camera 558 preferably includes one or more programmable DSLR cameras that provide pan, tilt, and zoom capabilities. Suitable DSLR cameras are available from Nikon Corp. and Canon, Inc. Stationary camera 558 may also include one or more Internet protocol (IP) cameras adapted for connecting to a telecommunications network, for example one utilizing the Internet Protocol communications protocol, such as the Internet. Suitable IP cameras are available from GeoVision, Inc. Arecont Vision provides suitable IP video cameras and associated software. It is also contemplated that stationary camera 558 is capable of directly coupling to a computing device via a relay or a communications channel employing serial and/or parallel communications methods. In addition, various filters, such as polarizing filters, may be

employed on stationary camera **558** for improving image quality. In an embodiment, stationary camera **558** is adapted for acquiring images of the exterior of the vehicle in a “zoomed out” perspective, while movable camera and light source **556** is adapted for acquiring images of the exterior of the vehicle in a “zoomed in” perspective.

(157) The warning light **559** is adapted for alerting users as well as bystanders of the operation of imaging apparatus **550** by flashing or blinking a light source, such as an LED. Warning light **559** is affixed to extension support member **552**. In one embodiment, warning light **559** flashes when imaging apparatus **550** is moving by self-locomotion. In another embodiment, warning light **559** flashes when camera support arm **555** pivots, extends, or moves in other respects. In another embodiment, warning light **559** may be used as a visual signal to a user that certain actions need to be taken with respect to imaging apparatus **550** or that imaging apparatus **550** has completed various portions of an image acquisition process.

(158) According to aspects of the invention, speaker **560** is adapted for alerting users as well as bystanders of the operation of imaging apparatus **550** by producing various sounds. Speaker **560** is affixed inside body support component **551** and utilizes openings in body support component **551** for sound to escape. In one embodiment, speaker **560** produces sounds when imaging apparatus **550** is moving by self-locomotion. In another embodiment, speaker **560** produces sounds when camera support arm **555** pivots, extends, or moves in other respects. In another embodiment, speaker **560** may be used as an audible signal to a human user that certain actions need to be taken with respect to imaging apparatus **550** or that imaging apparatus **550** has completed various portions of an image acquisition process.

(159) Referring further to FIG. 5D, wireless transceiver **561** is adapted for transmitting acquired images to computer **103**, server **104**, and/or database **105** via a communications medium, such as the Internet. Additionally, wireless transceiver **561** is adapted for receiving control signals generated by computer **103** or another computing device via a communications medium, such as the Internet. The control signals may provide information regarding the relative position of imaging apparatus **550** with respect to the vehicle, regarding movements of camera support arm **555**, and regarding the operation and manipulation of movable camera and light source **556** and stationary camera **558**. Wireless transceiver **561** is affixed to body support component **551** and has an antenna that extends from the top of body support component **551**. Wireless transceiver **561** may be any transceiver capable of receiving and/or transmitting communications signals. By way of example and not limitation, wireless transceiver **561** may operate according the IEEE 802.11 (WiFi) standard, the IEEE 802.15.1 (Bluetooth™) standard, may be a cellular network modem, or may be a laptop computer having wireless communications capabilities. A suitable cellular network modem is the 341U available from Netgear, Inc. Advantageously, wireless transceiver **561** allows imaging apparatus **550** to receive and send information without the need for wires and to operate in a variety of locations.

(160) The kill switch **562** is adapted for providing an easy means for a human user to stop the operation of imaging apparatus **550**. For example, kill switch **562** may be an emergency stop button that, when pressed, immediately stops any self-locomotion of imaging apparatus **550** or manipulation of camera support arm **555**. Advantageously, kill switch **562** allows a user to quickly stop any erroneous operations of imaging apparatus **550** and contributes to the ability of imaging apparatus **550** to operate safely around humans.

(161) The access door **563** is adapted for providing a means to easily access components of imaging apparatus **550** that are located inside body support component **551**. In one embodiment, access door **563** slides to create an opening in the surface of body support component **551**. Access door **563** may open by other means, such as hinges, latches, and the like. In another embodiment, access door **563** includes a locking mechanism to prevent unauthorized opening. Advantageously, access door **563** allows body support component **551** to conceal certain components of imaging apparatus **550** while still providing a means for a human to access the concealed components for

maintenance, troubleshooting, and the like.

(162) The wheels **564** and motor **568** are adapted for providing self-locomotive capabilities for imaging apparatus **550**. Motor **568** rotates an axle that is connected to wheels **564** such that motor **568** provides the necessary rotational torque to turn wheels **564**. Motor **568** and wheels **564** also provides means for positioning imaging apparatus **550** at various locations relative to the vehicle. In one embodiment, skid-steer tracks or the like are used instead of wheels **564** to move imaging apparatus **550** across a surface to position imaging apparatus **550** relative to the vehicle.

(163) In FIG. 5D, balance support **565** is adapted for helping to support the load of imaging apparatus **550** and keeping it stable. In an embodiment, balance support **565** may provide anti-tip capabilities and be referred to as an anti-tip support.

(164) According to aspects of the invention, processor **566** is adapted for controlling various components of imaging apparatus **550**. In one embodiment, processor **566** executes computer-executable instructions and generates electronic control signals to control vertical motion member **554**, movable camera and light source **556**, stationary camera **558**, warning light **559**, speaker **560**, wheels **564**, and motor **568** and send electronic signals via wireless transceiver **561**. Further, processor **566** may receive electronic signals from vertical motion member **554**, movable camera and light source **556**, stationary camera **558**, wireless transceiver **561**, and kill switch **562** in order to transform them into electronic control signals. In one embodiment, processor **566** includes an associated memory that stores computer-executable instructions and electronic data. In one embodiment, processor **566** receives control signals from computer **103** and transforms them into electronic control signals to manipulate respective components of imaging apparatus **550** accordingly. In another embodiment, processor **566** executes computer-executable instructions to generate electronic control signals to control various components of imaging apparatus **550**. In another embodiment, computing device **270** is an integrated circuit containing one or more processing units, a memory, an arithmetic logic unit, and/or a control unit.

(165) The power source **567** is adapted for providing electrical energy to power the various components of imaging apparatus **550**, including vertical motion member **554**, camera support arm **555**, movable camera and light source **556**, stationary camera **558**, warning light **559**, speaker **560**, wireless transceiver **561**, processor **566**, motor **568**, and sensors **569**. Power source **567** is affixed inside body support component **551**. In one embodiment, power source **567** is a battery that can be recharged or replaced, such as, by way of example and not limitation, an automotive battery. In another embodiment, power source **567** is a photovoltaic cell.

(166) The sensors **569** are adapted for providing information about the attributes of a surrounding environment of imaging apparatus **550**, including the subject being imaged. For example, sensors **569** may utilize ultrasonic, radar, sonar, or infrared propagation techniques to determine the distance from imaging apparatus **550**, or portions thereof, to another object. In an embodiment, sensors **569** aid in allowing imaging apparatus **550** to avoid obstacles. In another embodiment, sensors **569** allow imaging apparatus **550** to operate and acquire images without human involvement. Further, sensors **569** may provide measurement or dimension information about the subject being imaged.

(167) In operation of one embodiment, imaging apparatus **550** is positioned at a predefined starting location outside the footprint of an object to be imaged either by a user manually moving imaging apparatus **550** or by imaging apparatus **550** self-locomoting. For example, the predefined starting location may be a base station associated with imaging apparatus **550** that provides recharging of power source **567**. Once in the predefined location, processor **566** receives an electronic signal from computer **103** via wireless transceiver **561**. Computer-executable instructions executing on processor **566** generate electronic control signals to control various components of imaging apparatus **550**. For example, stationary camera **558** acquires images of the object, motor **568** powers wheels **564** to change the positioning of imaging apparatus **550**, warning light **559** flashes, speaker **560** generates audible sounds, vertical motion member **554** alters the vertical height of

camera support arm **555** and movable camera and light source **556**, camera support arm **555** pivots to extend movable camera and light source **556**, and movable camera and light source **556** provides sufficient lighting conditions and acquires images of the object. In an embodiment, imaging apparatus **550** is programmed to find its own way around the subject or vehicle being imaged such that it operates and acquires images in an autonomous manner.

(168) In one embodiment, imaging apparatus **550** comprises a body support component that provides structure to a plurality of components of the imaging apparatus and an enclosure for a plurality of components of the imaging apparatus. The imaging apparatus further comprising an extension support member extending from the body support component, a vertical support member affixed to a first portion of the extension support member, and a stationary camera affixed to a second portion of the extension support member. The imaging apparatus further comprising a vertical motion member affixed to the vertical support member, a camera support arm affixed to the vertical motion member at a first end, a movable camera and light source affixed to a second end of the camera support arm. The imaging apparatus further comprising a camera housing affixed to the body support component and adapted to receive the movable camera and light source. The imaging apparatus further comprising a warning light affixed to the extension support member, a speaker enclosed within the body support component, a wireless transceiver enclosed within the body support component, a kill switch affixed to the extension support member, a balance support affixed to the body support component, a processor enclosed within the body support component, a power source enclosed within the body support component, and a motor enclosed within the body support component. The imaging apparatus further comprising an access door affixed to the body support component and wheels mechanically linked to the motor.

(169) In addition to automobiles, various imaging apparatuses, methods, and/or imaging studios described herein are especially well suited to acquire exterior and/or interior images of an aircraft. In this embodiment, an imaging apparatus acquires images of the cockpit, cabin interior, fuselage exterior, top, bottom, wings, tail, and the like. For example, imaging apparatus **500** or imaging apparatus **550** may be used for this purpose. The acquired images are transmitted or transferred to computer **103**, server **104**, and/or database **105** according to embodiments of the invention.

(170) In another embodiment of the present invention, various imaging apparatuses, methods, and/or imaging studios described herein are suitable for acquiring images of real estate and associated structures and/or fixtures. For example, mobile image acquisition apparatus **200**, imaging apparatus **500**, and/or imaging apparatus **550** are suitable for this purpose. In another embodiment, one or more cameras as described herein are affixed to a tripod and placed in the middle of a room or space, indoors or outdoors, and images of the room or space are acquired as described herein. The imaging apparatus in this embodiment is capable of transmitting or transferring images to computer **103**, server **104**, and/or database **105**.

(171) FIGS. **6A-E** illustrate imaging studio **600**, which in one embodiment serves as the front-end image acquisition component **102** of FIG. **1**. The imaging studio **600** in the illustrated embodiment includes an optional vehicle transporter **602**, camera towers **604**, a camera boom **606**, and cameras **608**. The imaging studio **600** is configured to allow a vehicle to travel through locations **610**, **612**, **614**, **616**, **618**, and **620** (whether it moves independently or is carried through by the vehicle transporter **602**). In the embodiments of FIGS. **6A-E**, imaging studio **600** is adapted for acquiring images of a vehicle, but it is contemplated that imaging studio **600** may be used to acquire images of other objects. In one embodiment, imaging studio **600** is referred to as a finish line studio.

(172) As shown in the embodiment depicted in FIG. **6E**, the imaging studio can utilize a gantry structure with supports that function as camera towers **604** and a center beam that functions as camera boom **606**. In one embodiment, vehicle transporter **602** is used to carry a vehicle through imaging studio **600**. The vehicle transporter **602** is adapted for receiving a vehicle at a first location and moving it linearly past camera towers **604**, camera boom **606**, and cameras **608** to a second location. In the embodiment shown in FIG. **6A**, vehicle transporter **602** receives a vehicle at

location **610** and moves it past camera towers **604**, camera boom **606**, and cameras **608** to location **620**.

(173) In one embodiment, vehicle transporter **602** is a conveyer system that is seventy feet in length. Suitable conveyer systems include the DuraTrans® XD manufactured by Belanger, Inc. In the embodiment of FIG. **6B**, vehicle transporter **602** utilizes sensors to control the speed of vehicle movement and to signal vehicle location with respect to camera towers **604**, camera boom **606**, and cameras **608**. In this embodiment, vehicle transporter **602** receives a vehicle at location **610** and begins moving it toward camera towers **604**, camera boom **606**, and cameras **608**. When vehicle transporter **602** moves the vehicle into position **612**, a first sensor is tripped that initiates operation of cameras **608** that are used for a front 45-degree shot and a chassis shot, as further described below. When vehicle transporter **602** moves the vehicle into position **614**, a second sensor is tripped that ends operation of cameras **608** that are used for a front 45-degree shot and a chassis shot and initiates operation of cameras **608** that are used for a side shot, as further described below. When vehicle transporter **602** moves the vehicle into position **616**, a third sensor is tripped that ends operation of cameras **608** that are used for a side shot, as further described below. When vehicle transporter **602** moves the vehicle into position **618**, a fourth sensor is tripped that initiates operation of cameras **608** that are used for a rear 45-degree shot, as further described below. When vehicle transporter **602** moves the vehicle into position **620**, a fifth sensor is tripped that ends operation of cameras **608** that are used for a rear 45-degree shot, as further described below. FIG. **6E** provides a perspective view of the above embodiment.

(174) In another embodiment, imaging studio **600** does not include a vehicle transporter, and the vehicle travels freely from a first location and past camera towers **604**, camera boom **606**, and cameras **608** until it reaches a second location on the other side of imaging studio **600**. Sensors in imaging studio **600** measure the speed and position of the vehicle relative to camera towers **604**, camera boom **606**, and cameras **608**. When the vehicle moves to position **612**, a first sensor is tripped that initiates operation of cameras **608** that are used for a front 45-degree shot and a chassis shot, as further described below. When the vehicle moves to position **614**, a second sensor is tripped that ends operation of cameras **608** that are used for a front 45-degree shot and a chassis shot and initiates operation of cameras **608** that are used for a side shot, as further described below. When the vehicle moves to position **616**, a third sensor is tripped that ends operation of cameras **608** that are used for a side shot, as further described below. When the vehicle moves to position **618**, a fourth sensor is tripped that initiates operation of cameras **608** that are used for a rear 45-degree shot, as further described below. When the vehicle moves to position **620**, a fifth sensor is tripped that ends operation of cameras **608** that are used for a rear 45-degree shot, as further described below.

(175) In the embodiment of FIG. **6A**, camera towers **604** are adapted for providing a vertical structure to support and position at least one of cameras **608**. Preferably, camera towers **604** are ten feet in height, two feet wide, and are made of aluminum. In one embodiment, camera towers **604** are located twenty to thirty feet apart from each other, with vehicle transporter **602** located equidistant between them. In one embodiment, protection pipes surround camera towers **604** to help prevent camera towers **604** from being struck by a vehicle being imaged. FIG. **6C** illustrates a front view of camera towers **604** as viewed from location **610** along vehicle transporter **602**. FIG. **6D** illustrates a side view of camera towers **604** as viewed from a point between location **614** and location **616** along vehicle transporter **602**. FIG. **6E** illustrates a perspective view of camera towers **604**.

(176) Referring to FIG. **6C**, the camera boom **606** is adapted for providing a structure to support and position at least one of cameras **608** above vehicle transporter **602**. Preferably, camera boom **606** is affixed to the top of one of camera towers **604** at a height of ten feet and extends from that camera tower to a point above vehicle transporter **602**. In one embodiment, camera boom **606** is made of aluminum. FIG. **6C** illustrates a front view of camera boom **606** as viewed from location

610 along vehicle transporter **602**. An arch or applicator bar in an exterior rollover car wash system is suitable for adapting to support camera tower **604** and camera boom **606**. FIG. **6E** illustrates a perspective view of camera boom **606** functioning as the cross beam of a gantry structure.

(177) Referring further to FIG. **6A**, **6C**, and **6E**, cameras **608** are adapted for acquiring images of the exterior of the vehicle. In one embodiment, camera **608A** is located underneath the vehicle along the path of vehicle transporter **602** at a point before the vehicle reaches camera towers **604** and camera boom **606** and acquires images of the vehicle chassis or other portions of the vehicle that are visible from underneath. In one embodiment, camera **608A** is adapted for acquiring images in response to a first sensor signal when the vehicle is between location **612** and location **614**. Camera **608B** is affixed to camera boom **606** and is located at a point substantially ten feet above vehicle transporter **602**. In one embodiment, camera **608B** is adapted for acquiring images of the top of the vehicle in response to a second sensor signal when the vehicle is between location **614** and location **616**.

(178) Cameras **608C**, **608D**, and **608E** are affixed to camera towers **604**. Cameras **608C** are located eight to ten feet above vehicle transporter **602**. In one embodiment, cameras **608C** are adapted for acquiring images of the roof of the vehicle in response to a second sensor signal when the vehicle is between location **614** and location **616**. Cameras **608D** are located five feet above vehicle transporter **602**. In one embodiment, cameras **608D** are adapted for acquiring images of the sides of the vehicle in response to a second sensor signal when the vehicle is between location **614** and location **616**. Cameras **608E** are located two feet above vehicle transporter **602**. In one embodiment, cameras **608E** are adapted for acquiring images of the wheels of the vehicle in response to a second sensor signal when the vehicle is between location **614** and location **616**.

(179) Cameras **608F** are located between 5 and 10 feet above vehicle transporter **602**. In one embodiment, cameras **608F** are adapted for acquiring images of the front of the vehicle at a 45-degree angle in response to a first sensor signal when the vehicle is between location **612** and location **614**. In another embodiment, cameras **608F** are adapted for acquiring images of the rear of the vehicle at a 45-degree angle in response to a fourth sensor signal when the vehicle is between location **618** and location **620**. It is contemplated that the height of cameras **608A-F** may be altered from those described above to accommodate a vehicle with larger or smaller dimensions.

(180) In an embodiment, cameras **608** are adapted for simultaneously acquiring images of the exterior of a vehicle in both a “zoomed-out” perspective and a “zoomed-in” perspective. For example, cameras **608-F** may be adapted for acquiring images in the zoomed-out perspective, while cameras **608-D** may be adapted for acquiring images in the zoomed-in perspective.

(181) The cameras **608** preferably include one or more programmable DSLR cameras that provide pan, tilt, and zoom capabilities. Suitable DSLR cameras are available from Nikon Corp. and Canon, Inc. Cameras **608** may also be one or more Internet protocol (IP) cameras adapted for connecting to a telecommunications network, for example one utilizing the Internet Protocol communications protocol, such as the Internet. Suitable IP cameras are available from GeoVision, Inc. Arecont Vision provides suitable IP video cameras and associated software. In one embodiment, cameras **608** include video cameras capable of recording at 4K, or 4K2K, resolution. It is also contemplated that cameras **608** are capable of directly coupling to a computing device via a relay or a communications channel employing serial and/or parallel communications methods. In addition, various filters, such as polarizing filters, may be employed on cameras **608** for improving image quality. In one embodiment, cameras **608** transmit acquired images to computer **103**, server **104**, and/or database **105**, which can store, process, and/or mine the information for patterns, analogous values, and the like as described above.

(182) Beneficially, one embodiment of imaging studio **600** provides the ability for one or two users to acquire images of forty to sixty vehicles in one hour. In one embodiment, imaging studio **600** provides the ability to acquire video and/or as many as sixty images of each vehicle including images of the chassis, wheels, roof, hood, and trunk, 45-degree images, side view images, 360-

degree interior images. In one embodiment, imaging studio **600** is used in conjunction with imaging apparatus **500**, described herein. In another embodiment, imaging studio **600** is used in conjunction with imaging apparatus **550**, described herein. For example, imaging apparatus **500** and/or imaging apparatus **550** are configured to acquire images of the interior of a vehicle before or after transport through imaging studio **600**. In another embodiment, imaging apparatus **600** is used in conjunction with one or more handheld cameras.

(183) In another embodiment, cameras **608** utilize a processor to facilitate receiving and transmitting electronic control data and acquired image data. In an additional embodiment, cameras **608** may utilize computer **103**, which displays on a GUI a systematic process for acquiring images of an object and acquired images (e.g., ceiling shots, undercarriage shots, etc.). FIGS. 7A-7F illustrate an exemplary GUI for acquiring images and/or information and receiving commands from a user. Computer **103** may also display on a GUI stock images in a grayscale format for images of the object from perspectives that have yet to be acquired. In another embodiment, cameras **608** utilize an integrated circuit containing one or more processing units, a memory, an arithmetic logic unit, and/or a control unit to facilitate receiving and transmitting electronic control data and acquired image data.

(184) In one embodiment, a method for acquiring images for rendering a virtual vehicle showroom comprises connecting a vehicle to a vehicle transporter at a first predetermined location, transporting the vehicle along a path to a second predetermined location, and acquiring a plurality of images of the vehicle with a plurality of cameras while the vehicle is transported along the path. In another embodiment, the method further comprises transforming the acquired images into a stitched 360-degree panoramic image of the vehicle.

(185) According to further aspects of the invention, images acquired by a third-party provider and transmitted or transferred to computer **103** serves as the front-end image acquisition component **102** of FIG. 1. For example, a third-party provider acquires images using apparatuses, methods, and/or studios described herein and then transmits them to computer **103**, server **104**, and/or database **105** using a communications network or transfers them to computer **103**, server **104**, and/or database **105** via an external storage medium.

(186) FIGS. 8A-B illustrate an exemplary imaging apparatus **800**, which in one embodiment serves as the front-end image acquisition component **102** of FIG. 1. FIG. 8A provides a top perspective view, while FIG. 8B provides a bottom perspective view of the imaging apparatus **800**. The imaging apparatus **800** includes a hub **802**, arms **804**, and cameras **806**. In an embodiment, imaging apparatus **800** is capable of installation in a pre-existing structure. For example, imaging apparatus **800** may be installed in an automotive service bay. Advantageously, imaging apparatus **800** embodying aspects of the invention is capable of imaging a subject (e.g., a vehicle) in a period of about five seconds and creating an entry for the subject in an online inventory of subjects as further described herein. In the illustrated embodiment, a camera **806** is physically connected to each arm **804** and each arm **804** is physically connected to hub **802**. A camera **806** is also physically connected to hub **802** in the illustrated embodiment. In another embodiment, imaging apparatus **800** is used in conjunction with imaging apparatus **500**, described herein. In yet another embodiment, each camera **806** acquires images of the subject at a substantially simultaneous time. The acquired images may be stitched together by a processor executing processor-executable instructions to form a 360-degree image, as described herein.

(187) The hub **802** is adapted to support arms **804**. In an embodiment, hub **802** is an octagonal box comprised of a rigid material such as polyvinyl chloride (PVC), for example. The arms **804** are adapted to extend from hub **802** and support arm cameras **806** such that arm cameras **806** are positioned around the periphery of the subject being imaged. The embodiment illustrated in FIGS. 8A-B includes eight support arms **804** extending outwardly from octagonal hub **802** at increments of about forty-five degrees. In an embodiment, arms **804** comprise metallic conduit poles (e.g., electrical metallic tubing (EMT)) having a diameter of three-quarters of an inch ($\frac{3}{4}$ ") and a length

of ten feet (10').

(188) The cameras **806** are adapted for acquiring images of the subject. The cameras **806** preferably include one or more Internet protocol (IP) cameras adapted for connecting to a telecommunications network, for example one utilizing the Internet Protocol communications protocol, such as the Internet. Suitable IP cameras are available from GeoVision, Inc. Arecont Vision provides suitable IP video cameras and associated software. The cameras **806** may also include one or more DSLR cameras that provide pan, tilt, and zoom capabilities. Suitable DSLR cameras are available from Nikon Corp. and Canon, Inc. In another embodiment, cameras **806** are communicatively connected to computer **103**, receive electronic command signals from computer **103**, and transmit acquired image data to computer **103**. In another embodiment, cameras **806** are adapted for directly coupling to computer **103** via a relay or communications channel employing serial and/or parallel communications methods. In yet another embodiment, cameras **806** transmit acquired image data to server **104** and/or database **105**. In addition, various filters, such as polarizing filters, may be employed on cameras **806** for improving image quality. In an embodiment, physical media (e.g., wires) that communicatively and/or electrically connect to cameras **806** are enclosed within an inner cavity of arms **804**. One having skill in the art will understand that cameras **806** may also be communicatively and/or electrically connected to aspects of automated studio **100** via wireless channels.

(189) FIG. **9A** illustrates an exploded side view of aspects of imaging apparatus **800** and supports **818**. In the illustrated embodiment, imaging apparatus **800** further includes a non-bent portion **804-A** and a bent portion **804-B** of arm **804**, a connector sleeve **810**, a box **812**, a connector **814**, a cover **816**, and the supports **818**. The supports **818** comprise carabiners **820**, a turnbuckle **822**, a coupler **824**, and a clamp **826**. In an embodiment, imaging apparatus **800** is utilized in conjunction with a subject alignment pattern **828** affixed to a support surface **830**.

(190) In the embodiment illustrated in FIG. **9A**, a connector **814** is physically connected to hub **802** and an end of non-bent portion **804-A** of arm **804**. The connector **814** is preferably an EMT set screw connector having a diameter of three-quarters of an inch ($\frac{3}{4}$ "). At an opposite end, non-bent portion of arm **804-A** is physically connected to one end of connector sleeve **810**. An opposite end of connector sleeve **810** is physically connected to bent portion **804-B** of arm **804**. In this manner, connector sleeve **810** connects non-bent portion **804-A** to bent portion **804-B**. The connector sleeve **810** is preferably a conduit connector sleeve for accepting non-bent portion **804-A** and bent portion **804-B** each having a diameter of three-quarters of an inch ($\frac{3}{4}$ "). The connector sleeve **810** also preferably includes screws having a length of one-half of an inch ($\frac{1}{2}$ "). In an embodiment, bent portion of arm **804-B** includes an elbow of about ninety degrees and is cut to a desired length. At an end opposite the connection to connector sleeve **810**, bent portion of arm **804-B** is physically connected to another connector **814** which in turn is physically connected to box **812**. The box **812** is preferably a round conduit box having five holes each having a diameter of three-quarters of an inch ($\frac{3}{4}$ "). Each box **812** also has a cover **816**, which is preferably comprised of round non-metallic plastic. In an embodiment, cameras **806** physically connect to the cover **816** and/or the box **812**.

(191) Aspects of imaging apparatus **800** are connected to a support structure (e.g., a ceiling, a wall, etc.) via supports **818**. In the embodiment of FIG. **9A**, supports **818** comprise carabiners **820** having a first end and an opposite second end, a turnbuckle **820** having a first end and an opposite second end, and a coupler **824** having a first end and an opposite second end. Preferably, the first end of one of the carabiners **820** is physically connected to hub **802** and/or arm **804**. The carabiner **820** may be connected to arm **804** via the clamp **826** (e.g., pipe clamp). The second end of that carabiner **820** is connected to the first end of turnbuckle **820**. The second end of turnbuckle **820** is connected to the first end of coupler **824** and the second end of coupler **824** is connected to the first end of the second carabiner **820**. The second end of the second carabiner **820** is adapted to connect to the support structure. Preferably, carabiners **820** include three-eighths inch ($\frac{3}{8}$ ") coated spring link carabiners and turnbuckles **822** include those having five-sixteenths inch ($\frac{5}{16}$ ") by nine and

three-eighths inch ($9\text{-}\frac{3}{4}$ ") eyes and/or eye. The coupler **824** in the embodiment illustrated in FIG. **9A** is preferably a wire having a diameter of three-eighths inch ($\frac{3}{4}$ "). It will be understood by one having ordinary skill in the art that coupler **824** may also comprise chains, as further described herein. The clamps **826** are preferably three-quarters inch ($\frac{3}{4}$ ") steel conduit pipe clamps and pipe hangers.

(192) The subject alignment pattern **828**, is adapted for providing a means to consistently position subjects for imaging with respect to cameras **806**. The embodiment illustrated by FIGS. **9A-D** is preferably utilized to acquire images of a vehicle. In an embodiment, subject alignment pattern **828** comprises subject alignment patterns **430A-C** as described herein.

(193) FIG. **9B** illustrates an exploded top view of aspects of imaging apparatus **800**. FIG. **9C** illustrates a full side view of imaging apparatus **800**. As illustrated, arms **804** extend towards support surface **830** for a length l and a camera **806** connected to hub **802** is a height h above support surface **830**. In this embodiment, h is preferably ten feet ($10'$) and l is preferably cut to a desired length. FIG. **9D** illustrates a full top view of imaging apparatus **800**. The arms **804** extend a distance $d_{\text{sub.1}}$ from the center of hub **802**, a support **818** is connected to arm **804** at a distance $d_{\text{sub.2}}$ from the center of hub **802**, and connector sleeve **810** connects to non-bent portion of arm **804-A** to bent portion of arm **804-B** at a distance $d_{\text{sub.3}}$ from a near edge of hub **802**. The support **818** is also located a distance $d_{\text{sub.4}}$ from the extended centerline of hub **802** and from an adjacent arm **804**. A vertex of octagonal hub **802** to an edge of arm **804** is a distance $d_{\text{sub.5}}$. In the embodiment of imaging apparatus **800** illustrated by FIGS. **9A-D**, $d_{\text{sub.1}}$ is preferably fifteen feet, seven and one-quarter inches ($15'\text{-}7\frac{3}{4}"$), $d_{\text{sub.2}}$ is preferably fifteen feet ($15'$), $d_{\text{sub.3}}$ is preferably ten feet ($10'$), $d_{\text{sub.4}}$ is preferably ten feet, seven and one-quarter inches ($10'\text{-}7\frac{1}{4}"$), and $d_{\text{sub.5}}$ is preferably three and one-quarter inches ($3\frac{3}{4}"$).

(194) FIGS. **10A-D** illustrate another embodiment of imaging apparatus **800**. In this embodiment, aspects of imaging apparatus **800** are connected to a structure mount **832** via supports **818**. In an embodiment, structure mount **832** is preferably an EMT conduit pole having a length of ten feet ($10'$) and a diameter of three-quarters of an inch ($\frac{3}{4}"$). The structure mount **832** is physically attached at a fixed location to a structure (e.g., ceiling, wall, etc.) via bolts, screws, welding, or the like. In the illustrated embodiment, couplers **824** of supports **818** are preferably three-eighths inch ($\frac{3}{4}"$) chains. As shown by FIG. **10C**, bent portion of arms **804-B** extends toward support surface **830** a length l , which is preferably two feet, eight inches ($2'\text{-}8"$) in this embodiment. The height h is preferably ten feet ($10'$) in this embodiment.

(195) As illustrated in the top view of FIG. **10D**, arms **804** extend the distance d_i from the center of hub **802** and connector sleeve **810** connects non-bent portion of arm **804-A** to bent portion of arm **804-B** at the distance $d_{\text{sub.3}}$ from a near edge of hub **802**. A vertex of octagonal hub **802** to an edge of arm **804** is a distance $d_{\text{sub.5}}$. A structure mount **832** is a distance $d_{\text{sub.6}}$ from a near edge of octagonal hub **802**. A support **818** is a distance $d_{\text{sub.7}}$ from a near edge of octagonal hub **802** and a distance $d_{\text{sub.8}}$ from the extended centerline of hub **802**. The distance $d_{\text{sub.8}}$ is also the distance between adjacent arms **804** in this embodiment. Another structure mount **832** is a distance d_o from the center of hub **802**. In the embodiment of imaging apparatus **800** illustrated by FIGS. **10A-D**, $d_{\text{sub.1}}$ is preferably fifteen feet, seven and one-quarter inches ($15'\text{-}7\frac{1}{4}"$), $d_{\text{sub.2}}$ is preferably fifteen feet ($15'$), $d_{\text{sub.3}}$ is preferably ten feet ($10'$), $d_{\text{sub.5}}$ is preferably three and one-quarter inches ($3\frac{3}{4}"$), $d_{\text{sub.6}}$ is preferably three feet, eleven and three-quarters inches ($3'\text{-}11\frac{3}{4}"$), $d_{\text{sub.7}}$ is preferably five feet, ten inches ($5'\text{-}10"$), $d_{\text{sub.8}}$ is preferably four feet, six inches ($4'\text{-}6"$), and $d_{\text{sub.9}}$ is preferably two and three-quarters inches ($2\frac{3}{4}"$).

(196) FIGS. **11A-D** illustrate yet another embodiment of imaging apparatus **800**. In this embodiment, aspects of imaging apparatus **800** are connected via supports **818** to structure mounts **832** that are preferably zinc-plated slotted angle irons having a dimension of one and one-half inches ($1\frac{1}{2}"$) by sixty inches ($60"$). The structure mounts **832** are physically attached at fixed locations to a structure (e.g., ceiling, wall, etc.) via bolts, screws, welding, or the like. In the

illustrated embodiment, couplers **824** of supports **818** are preferably three-eighths inch ($\frac{3}{8}$ ") chains. As shown by FIG. **11C**, bent portion of arms **804-B** extends toward support surface **830** a length **l**, which is preferably two feet, eight inches (2' -8") in this embodiment. The height **h** is preferably ten feet (10') in this embodiment.

(197) As illustrated in the top view of FIG. **11D**, support **818** is a distance **d.sub.11** from the center of hub **802** and connector sleeve **810** connects non-bent portion of arm **804-A** to bent portion of arm **804-B** at a distance **d.sub.3** from a near edge of hub **802**. The supports **818** are a distance **d.sub.12** from connector sleeves **810**. The center of hub **802** is also a distance **d.sub.5** from the center of structure mounts **832** above hub **802**. The centers of holes in octagonal hub **802** to which arms **804** are connected are a distance **d.sub.10** apart from each other. In the embodiment of imaging apparatus **800** illustrated by FIGS. **11A-D**, **d.sub.3** is preferably ten feet (10'), **d.sub.5** is preferably three and one-quarter inches ($3\frac{1}{4}$ "), **d.sub.10** is preferably four and three-quarter inches ($4\frac{3}{4}$ "), **d.sub.11** is preferably thirteen feet (13'), and **d.sub.12** is preferably two feet, five and three-quarters inches (2'-5 $\frac{3}{4}$ ")

(198) In an embodiment, imaging apparatus **800** is installed in a photo bay having a ceiling and a floor. The installation process begins by determining the center of the photo bay and marking the ceiling at the center. The process continues by making four marks on the ceiling, each forty-five degrees From the bay centerline and four and three-quarters inches ($4\frac{3}{4}$ ") from the center mark. Six marks are then made on the ceiling, four of which are forty-five degrees From the bay centerline and two of which are along the bay centerline. All six of the marks are each thirteen feet (13') from the center mark. The process continues by installing eight ceiling mounts **832** (e.g., slotted angle irons) below each of the ten marks on the ceiling. Two of the ceiling mounts **832** will connect to four supports **818** connected to hub **802**. A carabiner **820** is attached to each of the ten ceiling mounts **832** and then a coupler **824** is attached to each of the ten carabiners **820**. In an embodiment, couplers **824** are chains that are cut to result in a lower end of the coupler being a distance above the floor. Preferably, a lower end of the coupler is ten feet, six inches (10'-6") from the floor. A carabiner **820** is then attached to the lower end of each chain, a turnbuckle **822** is attached at an upper end to each carabiner **820** and then another carabiner **820** is attached to a lower end of each turnbuckle **822**. The hub **802** is attached to four carabiners **820** at the center of the bay with the front of the hub facing toward the front of the bay. The turnbuckles **822** and the lengths of couplers **824** are adjusted until a lower surface of the hub **802** is a distance above the floor. Preferably, this distance is ten feet (10') above the floor. A non-bent portion of arm **804-A** is connected to a first end of bent portion of arm **804-B** via a connector sleeve **810**. A clamp **826** is attached to a bent portion of arm **804-B** at a distance from the center of connector sleeve **810**. Preferably this distance is two feet, five and three-quarters inches (2'-5 $\frac{3}{4}$ "). This step is repeated to make additional arms **804**. For example, the step is repeated five times to result in six arms **804**. All arms **804** are installed into connectors **814** to physically connect to hub **802**. The arms **804** are suspended by attaching the clamps **826** to the remaining carabiners **820**. The turnbuckles **822** and couplers **824** are adjusted until arms **804** are substantially horizontal (e.g., parallel to the floor). A connector **814**, a box **812**, and a cover **816** are then installed on a lower end of each bent portion of arm **804-B** (i.e., an end not connected to connector sleeve **810**). In an embodiment, the arms **804** are conduit poles through which wire is pulled from hub **802** to the lower end of each arm. The wiring is attached to cameras **806** which are then physically connected to box **812** and/or cover **816** of each arm **804**.

(199) When introducing elements of the present invention or the preferred embodiments(s) thereof, the articles "a", "an", "the" and "said" are intended to mean that there are one or more of the elements. The terms "comprising", "including" and "having" are intended to be inclusive and mean that there may be additional elements other than the listed elements.

(200) Having described aspects of the invention in detail, it will be apparent that modifications and variations are possible without departing from the scope of aspects of the invention as defined in

the appended claims. As various changes could be made in the above constructions, products, and methods without departing from the scope of aspects of the invention, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

Claims

1. A method for acquiring images of a plurality of vehicles, the plurality of vehicles including a first vehicle and a second vehicle, comprising: moving the first vehicle underneath an image acquisition system, the image acquisition system including at least six arms and at least six peripheral cameras, each of the at least six arms includes a downwardly extending portion that extends down towards the vehicle, each of the at least six arms arranged in a spaced-apart relationship relative to adjacent ones of the at least six arms, each of the at least six arms arranged in a fixed position such that the at least six arms are in the same fixed position when acquiring images of the first vehicle and the second vehicle, each of the at least six peripheral cameras affixed to a corresponding one of the downwardly extending portions of the at least six arms; stopping the first vehicle below the image acquisition system; and while the first vehicle is stationary, acquiring front image data of a front of the first vehicle from a first one of the at least six cameras; while the first vehicle is stationary, acquiring back image data of a back of the first vehicle from a second one of the at least six cameras; while the first vehicle is stationary, acquiring first side image data of a first side of the first vehicle from a third one and a fourth one of the at least six cameras; and while the first vehicle is stationary, acquiring second side image data of a second side of the first vehicle from a fifth one and a sixth one of the at least six cameras.
2. The method of claim 1, further comprising: while the first vehicle is stationary, acquiring bottom image data of a bottom of the first vehicle from a bottom camera positioned beneath the stationary first vehicle.
3. The method of claim 1, further comprising: placing a mobile image acquisition apparatus in an interior of the first vehicle, the mobile image acquisition apparatus configured to acquire images of the interior of the first vehicle.
4. The method of claim 1, further comprising: identifying a damage feature of an exterior of the first vehicle based on one or more of the front image data, the back image data, the first side image data, and the second side image data.
5. The method of claim 4, further comprising: classifying the damage feature as a dent or a scratch; and determining a dimension of the damage feature.
6. The method of claim 1, wherein the at least six arms are coupled to a hub, the at least six arms extending radially outwardly from the hub.
7. The method of claim 1, further comprising: transmitting an electronic file associated with the first vehicle by a processor communicatively coupled to the image acquisition system to a display configured to display the electronic file, the electronic file including the front image data, the back image data, the first side image data, and the second side image data.
8. The method of claim 1, further comprising: scanning, by a scanning device, indicia on the first vehicle that provides information about the first vehicle, the indicia includes a vehicle identification number (VIN) for the first vehicle.
9. The method of claim 8, further comprising: transmitting an electronic file associated with the first vehicle by a processor communicatively coupled to the image acquisition system, the electronic file including the front image data, the back image data, the first side image data, the second side image data, and the vehicle identification number (VIN) for the first vehicle.
10. The method of claim 1, further comprising: tilting, panning, or zooming one or more of the at least six cameras.
11. The method of claim 1, further comprising: after the first vehicle moves away from the image

acquisition system and while each of the at least six arms remain in the same fixed position as when the first vehicle was imaged, moving the second vehicle underneath the image acquisition system; while the second vehicle is stationary, acquiring front image data of a front of the second vehicle from the first one of the at least six cameras; while the second vehicle is stationary, acquiring back image data of a back of the second vehicle from the second one of the at least six cameras; while the second vehicle is stationary, acquiring first side image data of a first side of the second vehicle from the third one and the fourth one of the at least six cameras; and while the second vehicle is stationary, acquiring second side image data of a second side of the second vehicle from the fifth one and the sixth one of the at least six cameras.

12. A system for acquiring images of a plurality of vehicles, the plurality of vehicles including a first vehicle and a second vehicle, the first vehicle and the second vehicle being imaged sequentially such that the first vehicle is imaged by the system and moved away from the system and, thereafter, the second vehicle is then imaged by the system, the system comprising: at least six arms coupled to a structure located above the vehicle and extending downwardly in a fixed position, each of the at least six arms being arranged in a spaced-apart relationship relative to adjacent ones of the at least six arms, each of the at least six arms arranged in a fixed position such that the at least six arms remain in the same fixed positions when acquiring images of the first vehicle and the second vehicle; a front camera affixed to a first one of the at least six arms, the front camera configured to acquire front image data of a front of the first vehicle and the second vehicle; a rear camera affixed to a second one of the at least six arms, the rear camera configured to acquire rear image data of a rear of the first vehicle and the second vehicle; two first side cameras affixed, respectively, to a third one and a fourth one of the at least six arms, the two first side cameras configured to acquire first side image data of a first side of the first vehicle and the second vehicle; two second side cameras affixed, respectively, to a fifth one and a sixth one of the at least six arms, the two second side cameras configured to acquire second side image data of a second side of the first vehicle and the second vehicle; wherein the front camera, the rear camera, the two first side cameras, and the two second side cameras acquire the front image data, the rear image data, the first side image data, and the second side image data of the first vehicle and the second vehicle when the first vehicle and the second vehicle are stationary.

13. The system of claim 12, further comprising a mobile image acquisition apparatus configured to acquire images of the interior of the first vehicle and the second vehicle.

14. The system of claim 12, further comprising a hub, wherein the at least six arms are coupled to the hub and extend radially outwardly from the hub.

15. The system of claim 12, further comprising a bottom camera positioned beneath the stationary first vehicle and the second vehicle and configured to acquire bottom image data of a bottom of the first vehicle and the second vehicle.

16. The system of claim 13, further comprising a top camera positioned above the stationary first vehicle and the second vehicle and configured to acquire top image data of a top of the first vehicle and the second vehicle.

17. The system of claim 12, further comprising a scanning device, the scanning device configured to scan vehicle identification numbers (VIN) of the first vehicle and the second vehicle.

18. The system of claim 12, further comprising a processor communicatively coupled to the front camera, the rear camera, the two first side cameras, and the two second side cameras, the processor configured to receive the front image data, the rear image data, the first side image data, and the second side image data of the first vehicle and the second vehicle and transmit an electronic file associated with the first vehicle and the second vehicle, the electronic file including the image data of the first vehicle and the second vehicle.

19. The system of claim 18, further including a server communicatively coupled to the processor, the server configured to examine the front image data, the rear image data, the first side image data, and the second side image data of the first vehicle and the second vehicle and determine whether a

damage feature is present on the first vehicle and the second vehicle.

20. The system of claim 19, wherein the server is further configured to determine a dimension of the damage feature.

21. The system of claim 20, wherein the electronic file further includes the dimension of the damage feature.

22. A method for acquiring images of a vehicle, comprising: moving the vehicle underneath an image acquisition system, the image acquisition system including at least six arms and at least six peripheral cameras, each of the at least six arms includes a downwardly extending portion that extends down towards the vehicle, each of the at least six arms are coupled to a hub and extend radially outwardly from the hub, each of the at least six arm is arranged in a spaced-apart relationship relative to adjacent ones of the at least six arms, each of the at least six peripheral cameras affixed to a corresponding one of the downwardly extending portions of the at least six arms; stopping the vehicle below the image acquisition system; and while the vehicle is stationary, acquiring front image data of a front of the vehicle from a first one of the at least six cameras; while the vehicle is stationary, acquiring back image data of a back of the vehicle from a second one of the at least six cameras; while the vehicle is stationary, acquiring first side image data of a first side of the vehicle from a third one and a fourth one of the at least six cameras; and while the vehicle is stationary, acquiring second side image data of a second side of the vehicle from a fifth one and a sixth one of the at least six cameras.
